

Module 8

Grid-Following and Grid-Forming Converters (Chaps. 7-8-9 of Yazdani's Book)

Most Important Topics and Concepts

- Different methods for controlling grid-connected or grid-imposed frequency VSC systems
 - Modeling and control in $\alpha\beta$ - and dq -coordinates
 - Controlling active and reactive power
 - Synchronization with the ac grid, design of PLL
 - Dc-side voltage control
 - Modeling and control of grid-forming converters
- Phase-locked loop*

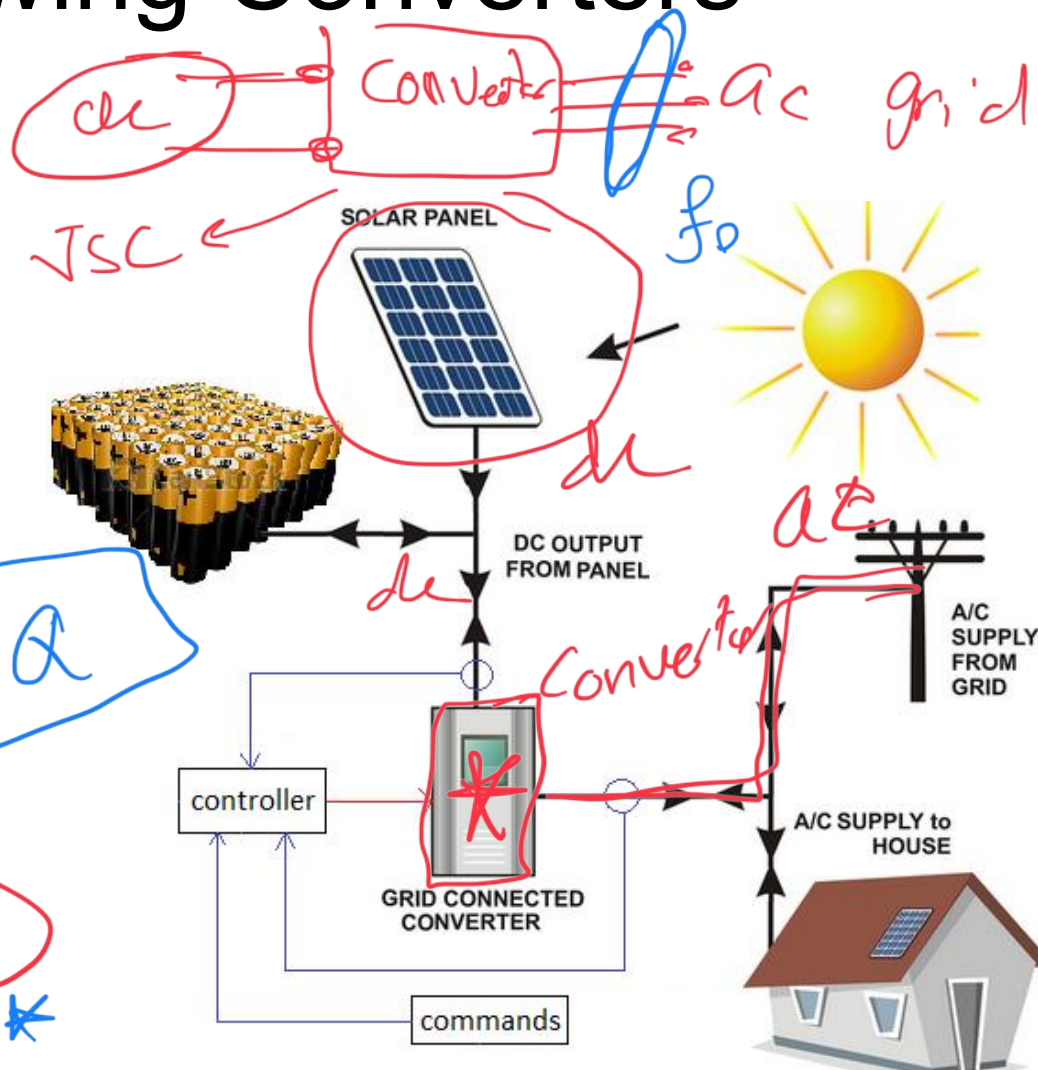
Grid-Following Converters

□ What we know:

- ✓ DC-side may be a DC source or another power-electronic system
- ✓ Frequency is relatively constant and imposed by the AC grid

□ What we want:

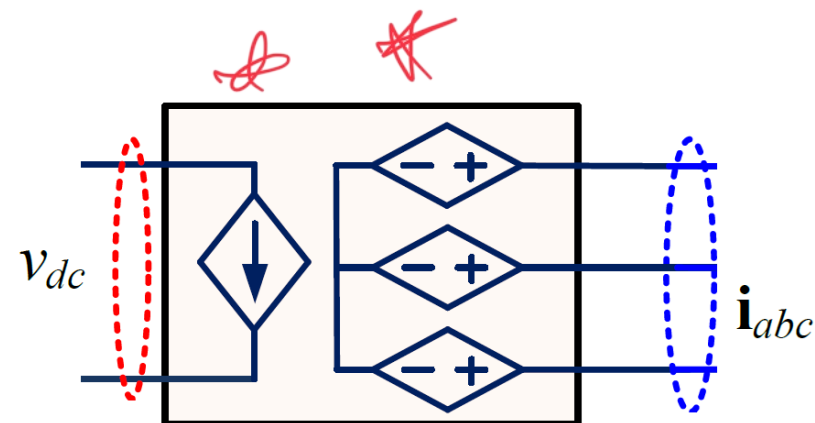
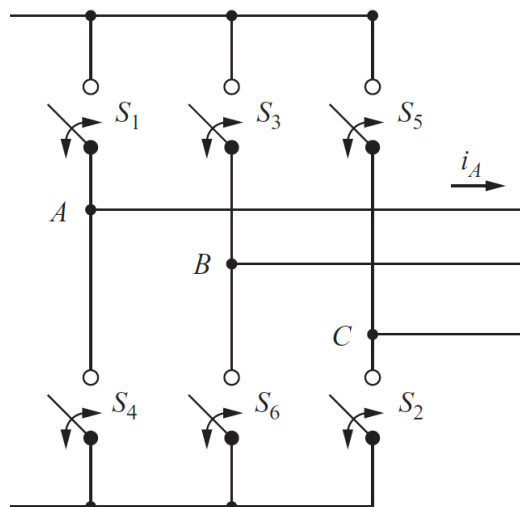
- ✓ VSC exchanges real/reactive power with AC system at PCC
- ✓ Control real/reactive power
(dc side connected to a dc voltage source)
- ✓ Control DC bus voltage
(dc side connected to a dc power source)



Three-Phase PWM Converter

❑ **Objective:** Producing 3-phase AC outputs from a DC source with PWM

❖ **3-Phase PWM Inverter:**

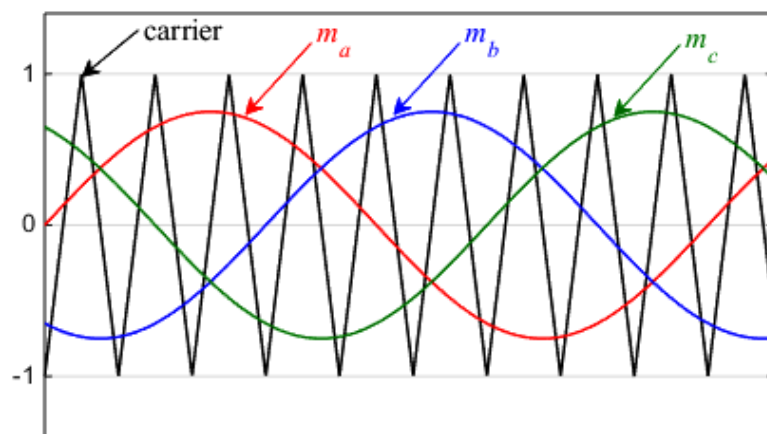


✓ Fundamental-frequency components of ac voltages:

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} m_a \\ m_b \\ m_c \end{bmatrix} \times \frac{v_{dc}}{2}$$

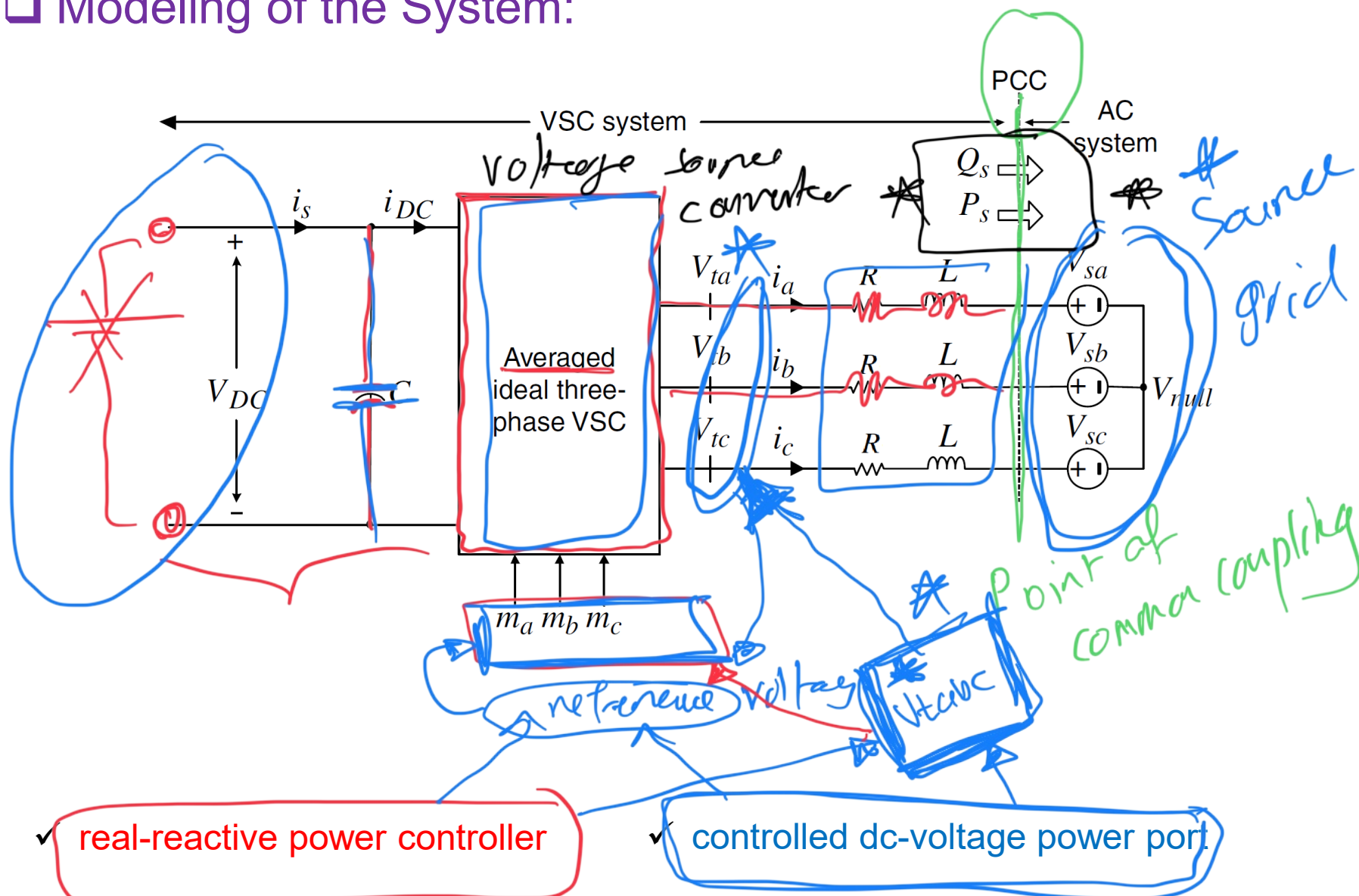
✓ Average-value of dc current:

$$i_{dc} = \frac{1}{2} \times (m_a i_a + m_b i_b + m_c i_c)$$



Grid-Following Converters

□ Modeling of the System:



- ❑ Real-Reactive Power Control:

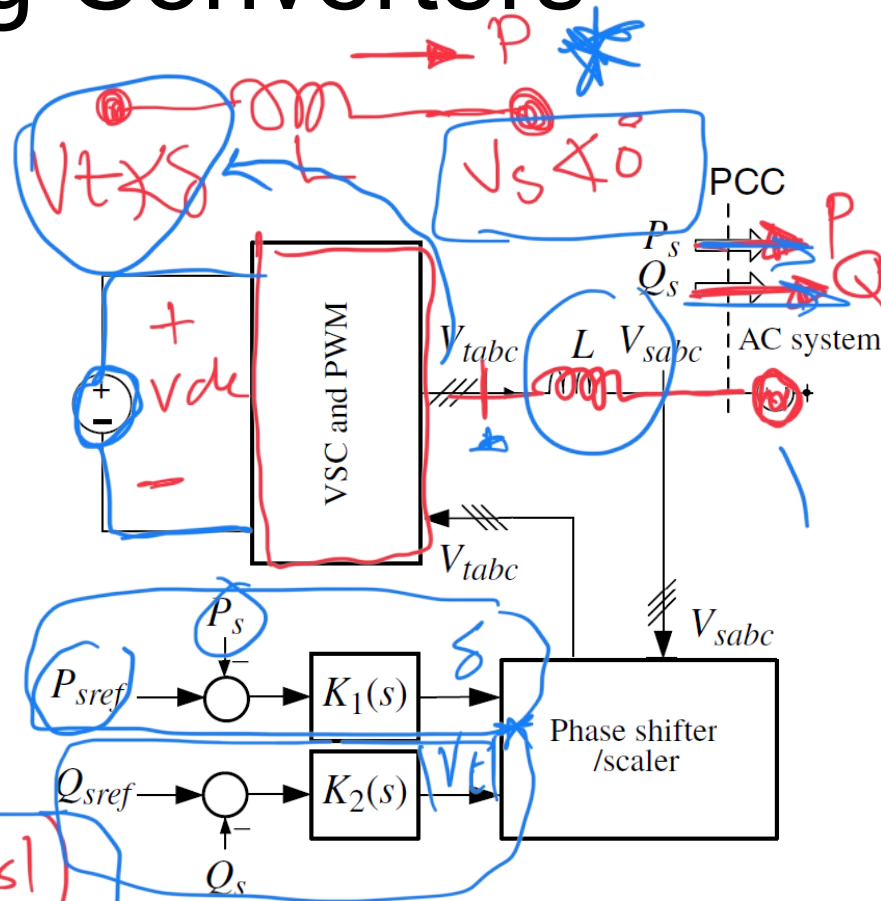
❖ Voltage-Mode Control:

$$Q_s = 3 \times \frac{|V_t| \times |V_s|}{X} \left[\overbrace{\cos(\delta)}^{\text{Power Factor}} - \underbrace{\frac{|V_s|}{|V_t|}}_{\text{Voltage Regulation}} \right]$$

small δ

$$P_S \propto S^*$$

$$Q_s \propto (|v_E| - |v_s|)$$



- ✓ P_s and Q_s may be controlled by respectively controlling the Phase Angle and Amplitude of VSC terminal voltages with respect to PCC voltages
- ✓ Simple technique with low number of control loops

- ✗ No control loop dedicated to currents!
- ✗ Currents can undergo large excursions (e.g., during faults or rapid change of commands)

Grid-Following Converters

□ Real-Reactive Power Control:

❖ Current-Mode Control:

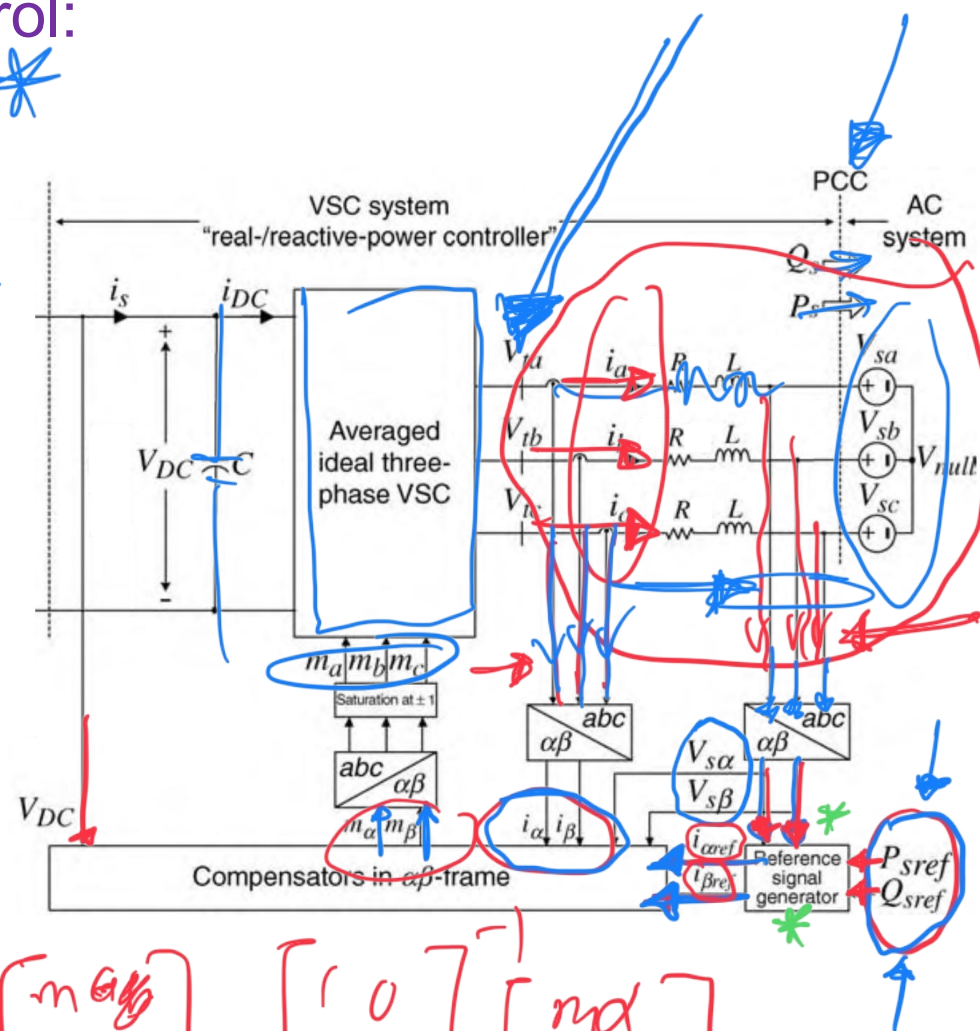
- ✓ P_s and Q_s are controlled by controlling the Phase Angle and Amplitude of VSC line currents with respect to PCC voltages
- ✓ VSC is protected against over currents because of tight current regulation
- ✓ Widely utilized for control of VSCs in power systems



- ✓ P_s and Q_s are controlled by the line current components i_α and i_β



- ✓ The reference currents $i_{\alpha ref}$ and $i_{\beta ref}$ are limited by saturation blocks



$$\begin{bmatrix} m_a \\ m_b \\ m_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1/2 & 1/\sqrt{3} \\ \sqrt{3}/2 & 1/2 \end{bmatrix} \begin{bmatrix} m_\alpha \\ m_\beta \end{bmatrix}$$

Grid-Following Converters

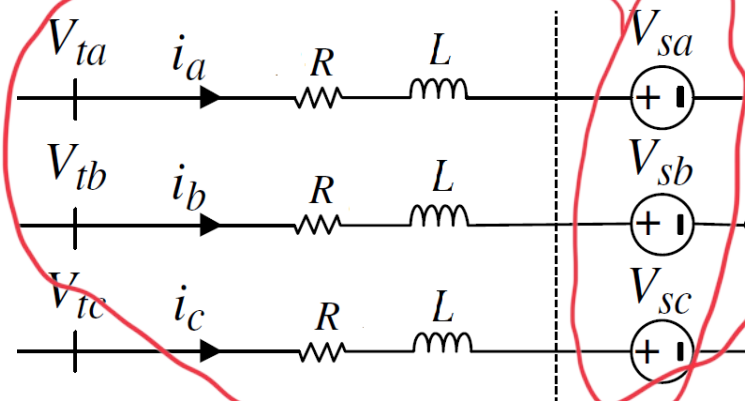
□ Dynamic Model of System:

➤ Voltages of the AC grid:

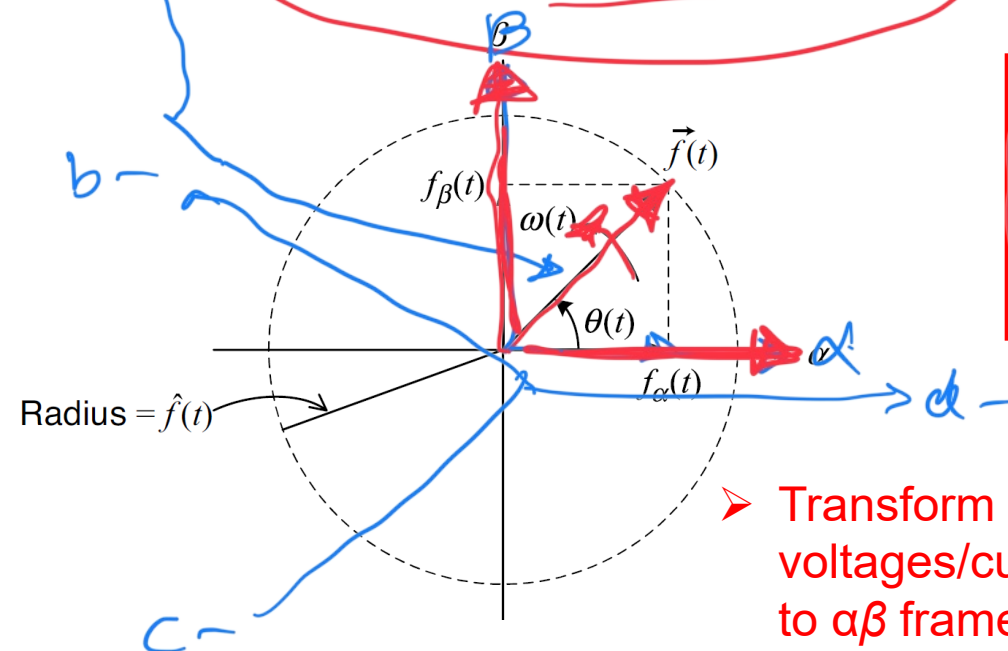
$$v_{sa}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0)$$

$$v_{sb}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0 - 2\pi/3)$$

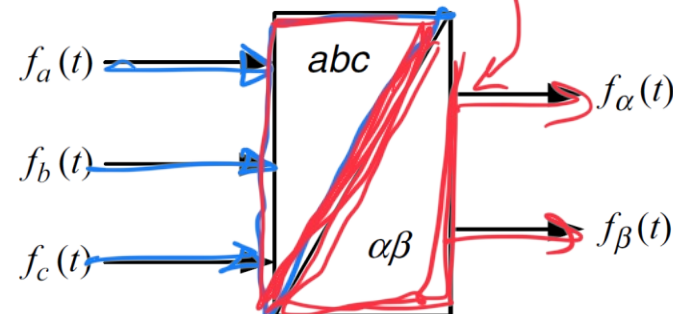
$$v_{sc}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0 + 2\pi/3)$$



$$\begin{bmatrix} f_\alpha(t) \\ f_\beta(t) \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} f_a(t) \\ f_b(t) \\ f_c(t) \end{bmatrix}$$



➤ Transform the *abc* voltages/currents to $\alpha\beta$ frame



Grid-Following Converters

Dynamic Model of System:

- Active/Reactive Power Delivered to the AC grid:

$$\begin{cases} P_s = \frac{3}{2} [v_{s\alpha} i_\alpha + v_{s\beta} i_\beta] \\ Q_s = \frac{3}{2} [v_{s\beta} i_\alpha - v_{s\alpha} i_\beta] \end{cases}$$

- What can be controller to control P and Q ?

$$i_{\alpha ref} = \frac{2}{3} \left[\frac{v_{s\alpha}}{v_{s\alpha}^2 + v_{s\beta}^2} P_{sref} + \frac{v_{s\beta}}{v_{s\alpha}^2 + v_{s\beta}^2} Q_{sref} \right]$$

$$i_{\beta ref} = \frac{2}{3} \left[\frac{v_{s\beta}}{v_{s\alpha}^2 + v_{s\beta}^2} P_{sref} - \frac{v_{s\alpha}}{v_{s\alpha}^2 + v_{s\beta}^2} Q_{sref} \right]$$

if current controller ensures

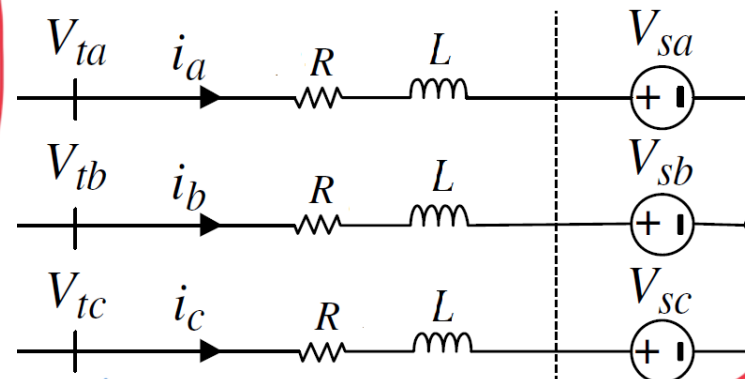
$$i_\alpha \approx i_{\alpha ref} \quad i_\beta \approx i_{\beta ref}$$

$$P_s \approx P_{sref} \quad Q_s \approx Q_{sref}$$

$$P_s v_{s\alpha} + Q_s v_{s\beta} =$$

$$\frac{3}{2} (v_{s\alpha}^2 + v_{s\beta}^2) i_\alpha$$

$$\xrightarrow{P_s} Q_s$$



$$\begin{aligned} P_s \cdot v_{s\alpha} &= \frac{3}{2} [v_{s\alpha}^2 i_\alpha + v_{s\beta} v_{s\alpha} i_\beta] \\ Q_s \cdot v_{s\beta} &= \frac{3}{2} [v_{s\beta}^2 i_\alpha - v_{s\alpha} v_{s\beta} i_\beta] \end{aligned}$$

Grid-Following Converters

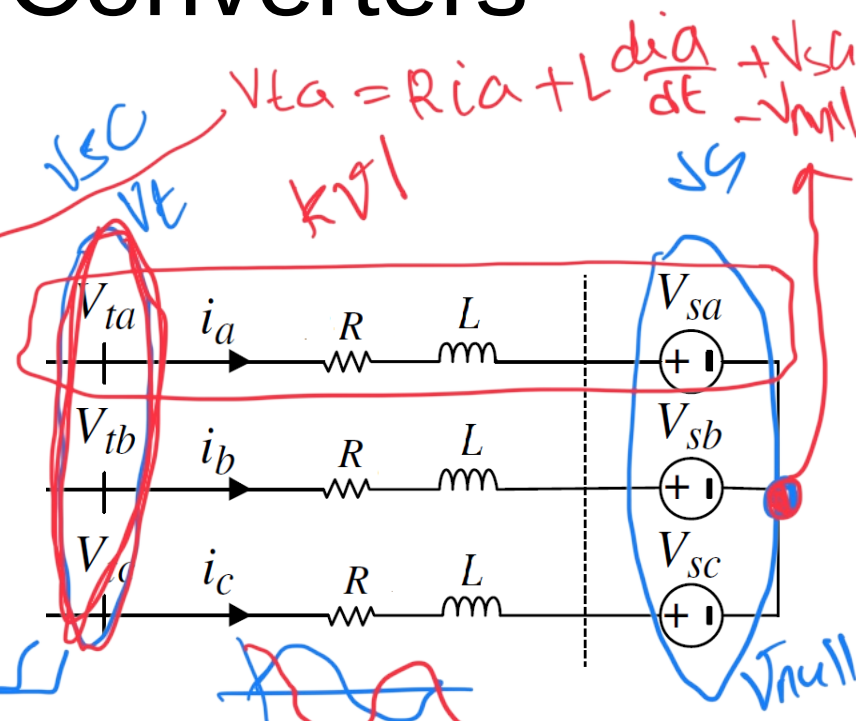
□ Dynamic Model of System:

➤ Dynamics of AC-side system:

$$L \frac{di_a}{dt} = -Ri_a + v_{ta} - v_{sa} - v_{null}$$

$$L \frac{di_b}{dt} = -Ri_b + v_{tb} - v_{sb} - v_{null}$$

$$L \frac{di_c}{dt} = -Ri_c + v_{tc} - v_{sc} - v_{null}$$



➤ Transform the abc voltages/currents to $\alpha\beta$ frame

$$v_{t\alpha} = \frac{V_{DC}}{2} m_{\alpha}$$

$$v_{t\beta} = \frac{V_{DC}}{2} m_{\beta}$$

$$L \frac{di_{\alpha}}{dt} = -Ri_{\alpha} + v_{t\alpha} - v_{s\alpha}$$

$$L \frac{di_{\beta}}{dt} = -Ri_{\beta} + v_{t\beta} - v_{s\beta}$$

$$L \frac{di_{\alpha}}{dt} = -Ri_{\alpha} + \frac{V_{DC}}{2} m_{\alpha} - v_{s\alpha}$$

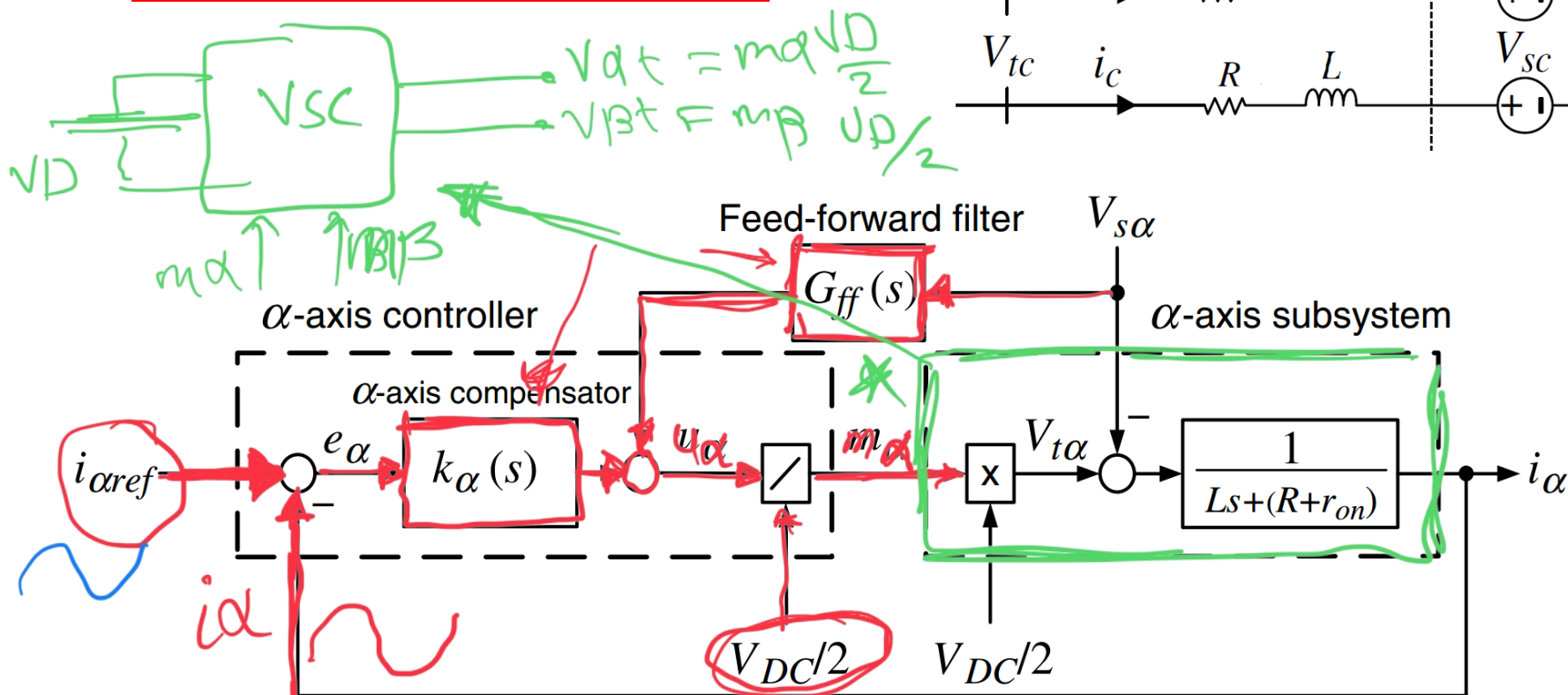
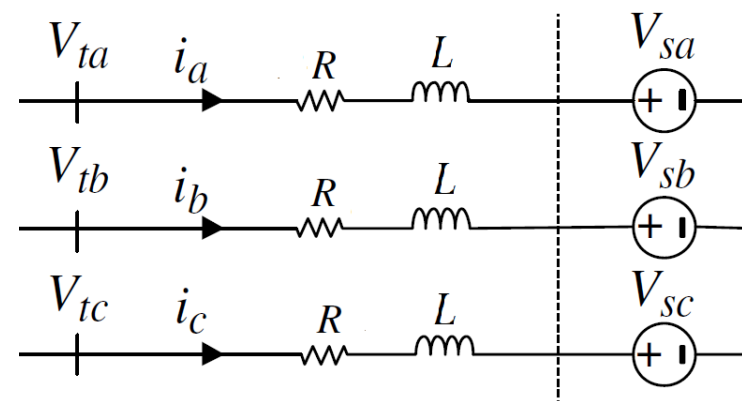
$$L \frac{di_{\beta}}{dt} = -Ri_{\beta} + \frac{V_{DC}}{2} m_{\beta} - v_{s\beta}$$

Grid-Following Converters

Dynamic Model of System:

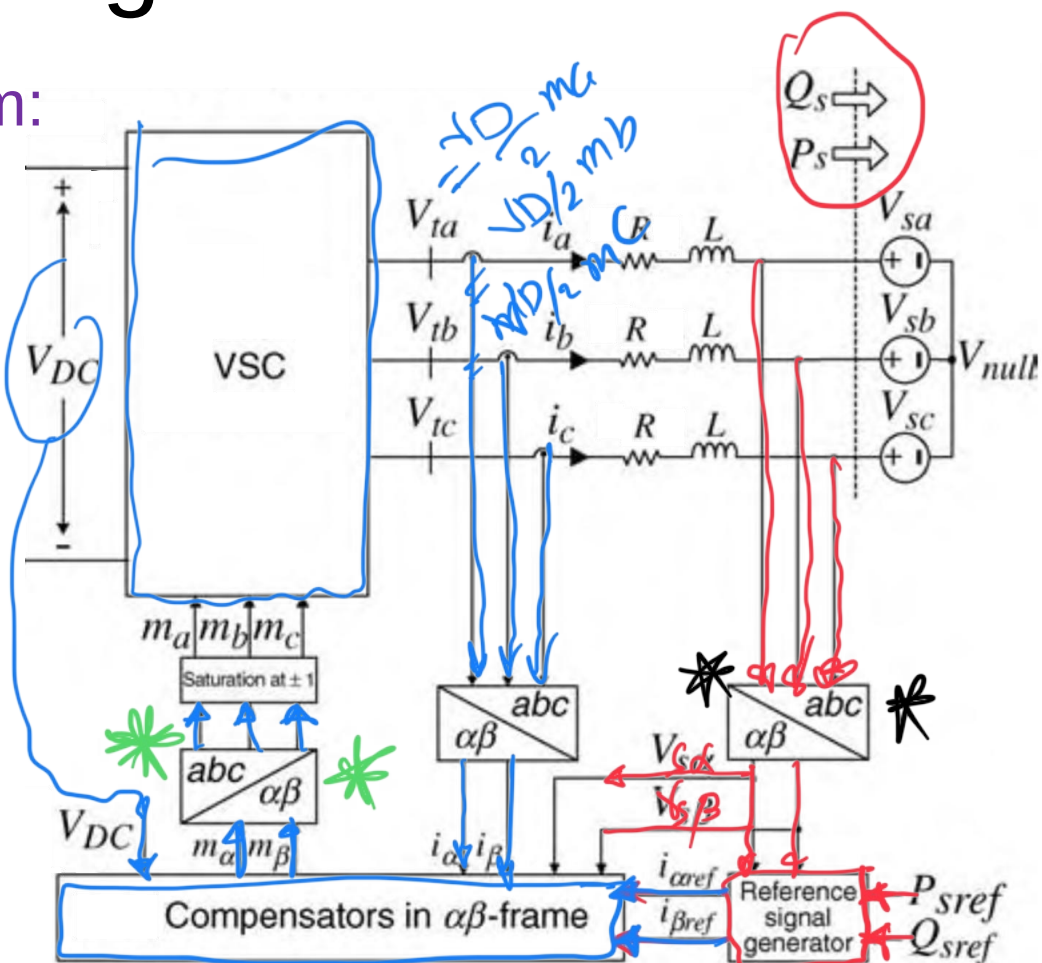
Current-Controllers:

$$L \frac{di_{\alpha}}{dt} = -Ri_{\alpha} + \frac{V_{DC}}{2} m_{\alpha} - v_{s\alpha}$$



❑ Dynamic Model of System:

$$\begin{bmatrix} f_a \\ f_b \\ f_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \\ -1/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} f_\alpha \\ f_\beta \end{bmatrix}$$



$$i_{\alpha ref} = \frac{2}{3} \left[\frac{v_{s\alpha}}{v_{s\alpha}^2 + v_{s\beta}^2} P_{sref} + \frac{v_{s\beta}}{v_{s\alpha}^2 + v_{s\beta}^2} Q_{sref} \right] \quad i_{\beta ref} = \frac{2}{3} \left[\frac{v_{s\beta}}{v_{s\alpha}^2 + v_{s\beta}^2} P_{sref} - \frac{v_{s\alpha}}{v_{s\alpha}^2 + v_{s\beta}^2} Q_{sref} \right]$$

Grid-Following Converters

□ Transient Study and with PQ Control:

➤ Example 7.1:

$$R = 1.63 \text{ m}\Omega \quad L = 100 \text{ }\mu\text{H}$$

$$V_s^{LL} = 480 \text{ V} \quad \omega_0 = 377 \text{ rad/s}$$

$$V_{DC} = 1450 \text{ V} \quad f_{sw} = 3420 \text{ Hz}$$

$$G_{ff}(s) = 1/(8 \times 10^{-6} s + 1)$$

$$k_\alpha(s) = k_\beta(s) = 1258 \left(\frac{s + 16.34}{s^2 + 377^2} \right) \left(\frac{s + 966}{s + 5633} \right) \left(\frac{s + 2}{s + 0.05} \right)$$

gating
pulses
blocked
controller
inactive

gates
unblocked
 $P_{sref} = 0$
 $Q_{sref} = 0$

P_{sref}

1 MW

P_{sref}

1 MW

-1 MW

Q_{sref}

0.5 MVar

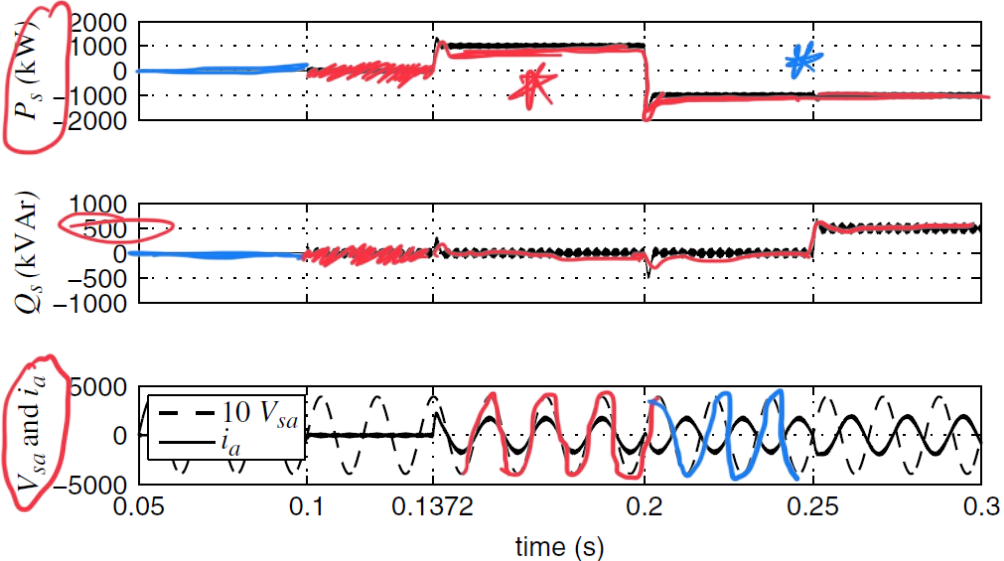
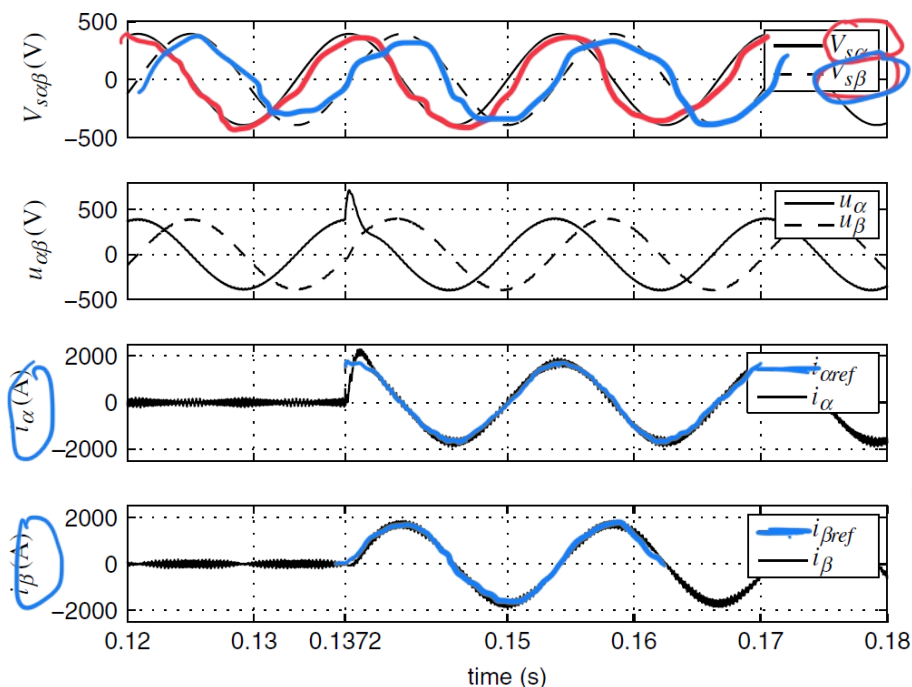
$t=0$

$t=0.1 \text{ s}$

$t=0.1372 \text{ s}$

$t=0.20 \text{ s}$

$t=0.25 \text{ s}$



Grid-Following Converters

Space Phasor in qd -Frame:

- ✓ Consider the space phasor of any 3-phase variables:

$$\vec{f}(t) = f_\alpha(t) + jf_\beta(t) =$$

$$\frac{2}{3} \left[e^{j0} f_a(t) + e^{j2\pi/3} f_b(t) + e^{-j2\pi/3} f_c(t) \right]$$

$$\vec{f}(t) = \hat{f} e^{j\theta(t)} = \hat{f} e^{j(\omega_0 t + \theta_0)}$$

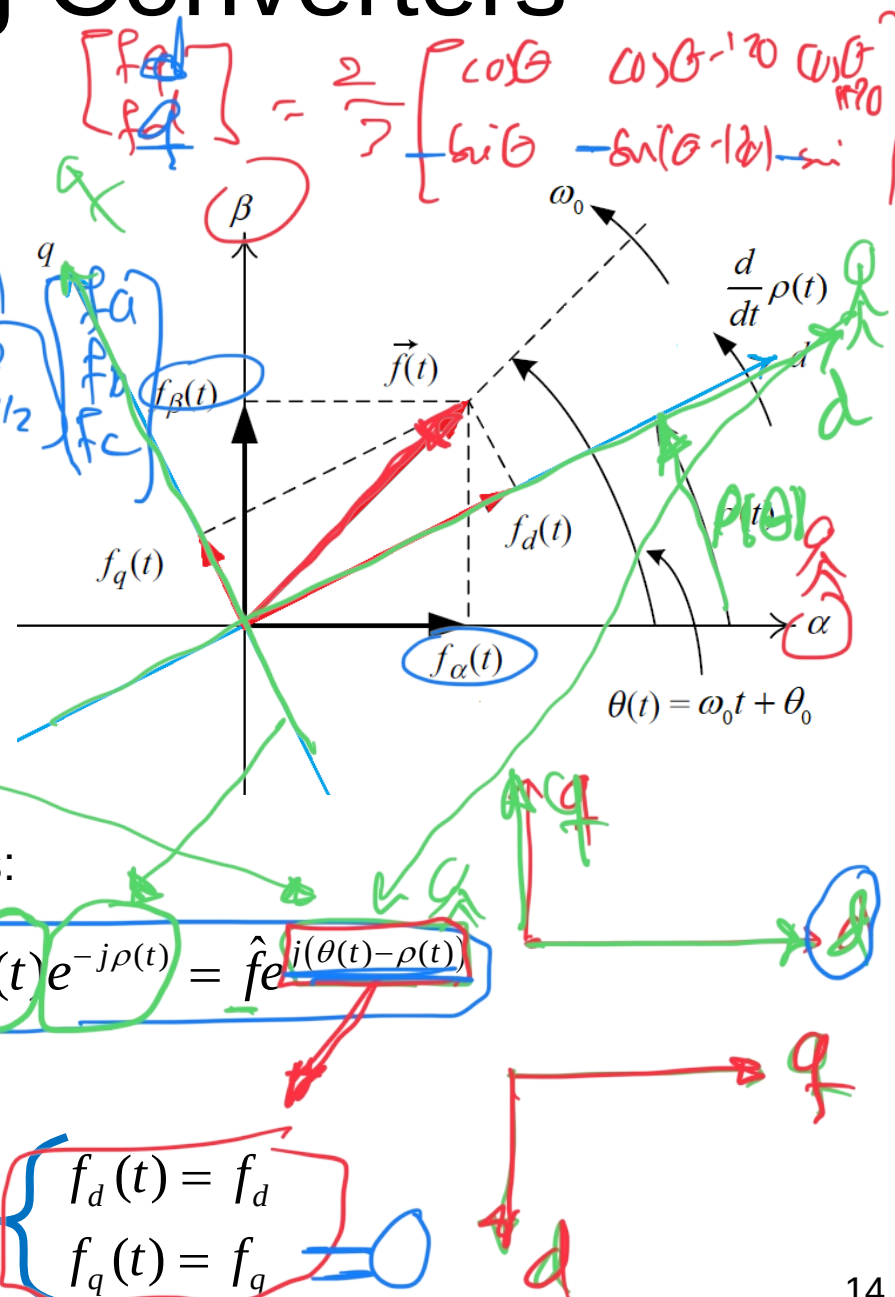
- ✓ The phasor in dq -frame is expressed as:

$$f_{dq}(t) = \underline{f_d(t)} + j\underline{f_q(t)} = \vec{f}(t) e^{-j\rho(t)} = \hat{f} e^{j(\theta(t) - \rho(t))}$$

- ✓ Angle $\rho(t)$ can be chosen arbitrarily!

$$\frac{d}{dt} \rho(t) = \frac{d}{dt} \theta(t) = \omega_0$$

$$\begin{cases} f_d(t) = f_d \\ f_q(t) = f_q \end{cases}$$



Grid-Following Converters

Real-Reactive Power Control in dq frame:

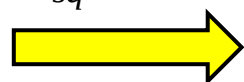
✓ Real and reactive power in dq -frame:

$$P_s = \frac{3}{2} (v_{sd} i_d + v_{sq} i_q)$$

$$Q_s = \frac{3}{2} (v_{sq} i_d - v_{sd} i_q)$$

$$v_{sq} = 0$$

PLL in steady-state



$$P_s = \frac{3}{2} v_{sd} i_d$$

$$Q_s = -\frac{3}{2} v_{sd} i_q$$

✓ P_s and Q_s are controlled by i_d and i_q respectively!

$$i_{dref} = \frac{2}{3v_{sd}} \times P_{sref}$$

$$i_{qref} = \frac{-2}{3v_{sd}} \times Q_{sref}$$

if current controller ensures

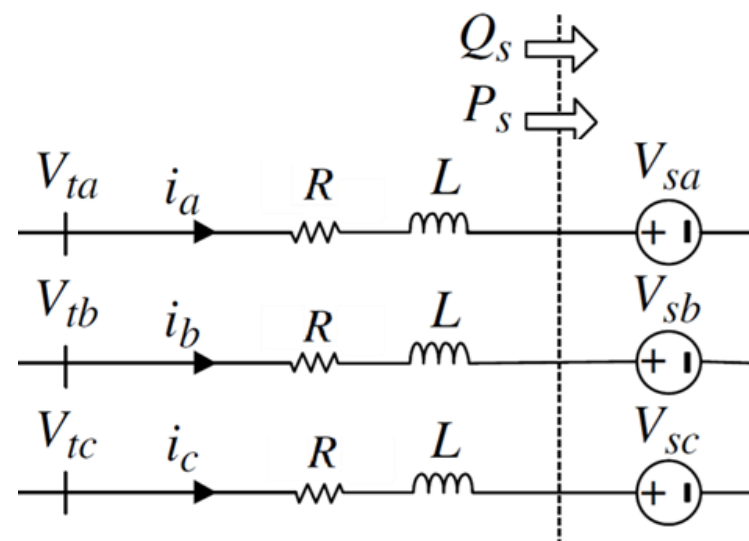
$$i_\alpha \approx i_{\alpha ref}$$

$$i_\beta \approx i_{\beta ref}$$



$$P_s \approx P_{sref}$$

$$Q_s \approx Q_{sref}$$

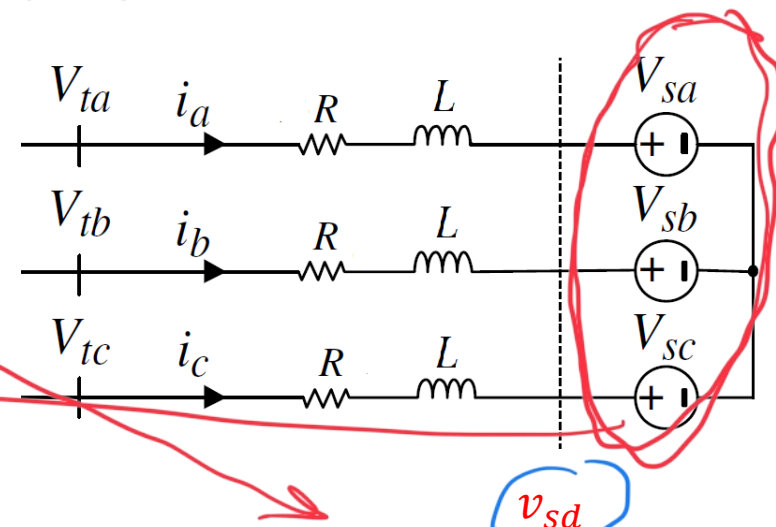


Grid-Following Converters

□ Dynamic Model of System in dq frame:

➤ Voltages of the AC grid:

$$\begin{cases} v_{sa}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0) \\ v_{sb}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0 - 2\pi/3) \\ v_{sc}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0 + 2\pi/3) \end{cases}$$



➤ Let $\frac{d}{dt} \rho(t) = \omega$ ➡

$$\begin{cases} L \frac{di_d}{dt} = -Ri_d + L\omega i_q + v_{td} - \hat{V}_s \cos(\omega_0 t + \theta_0 - \rho) \\ L \frac{di_q}{dt} = -Ri_q - L\omega i_d + v_{tq} - \hat{V}_s \sin(\omega_0 t + \theta_0 - \rho) \end{cases}$$

How to ensure this?

$\rho = \omega_0 t + \theta_0$ ➡

$$\begin{cases} v_{sd} = \hat{V}_s \\ v_{sq} = 0 \end{cases}$$

$v_{sq} = \hat{V}_s \sin(\omega_0 t + \theta_0 - \rho)$ ➡

✓ devise a mechanism to ensure ($v_{sq} = 0$)

Phase-Locked Loop (PLL)

Grid-Following Converters

□ Dynamic Model of System in dq frame:

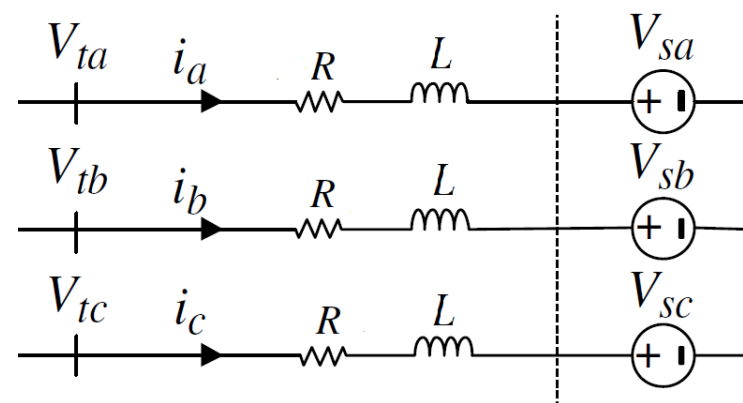
➤ Dynamics of AC-side system:

$$\begin{cases} L \frac{di_d}{dt} = L\omega_0 i_q - Ri_d + v_{td} - v_{sd} \\ L \frac{di_q}{dt} = -L\omega_0 i_d - Ri_q + v_{tq} - v_{sq} \end{cases}$$

coupled!

where

$$v_{td} = \frac{V_{DC}}{2} m_d \quad v_{tq} = \frac{V_{DC}}{2} m_q$$



✓ Introduce two intermediate control variables u_d and u_q so that:

$$\begin{cases} m_d = \frac{2}{V_{DC}} (u_d - L\omega_0 i_q + v_{sd}) \\ m_q = \frac{2}{V_{DC}} (u_q + L\omega_0 i_d + v_{sq}) \end{cases} \Rightarrow \begin{cases} L \frac{di_d}{dt} = -Ri_d + u_d \\ L \frac{di_q}{dt} = -Ri_q + u_q \end{cases} \Rightarrow \text{two decoupled linear first-order systems}$$

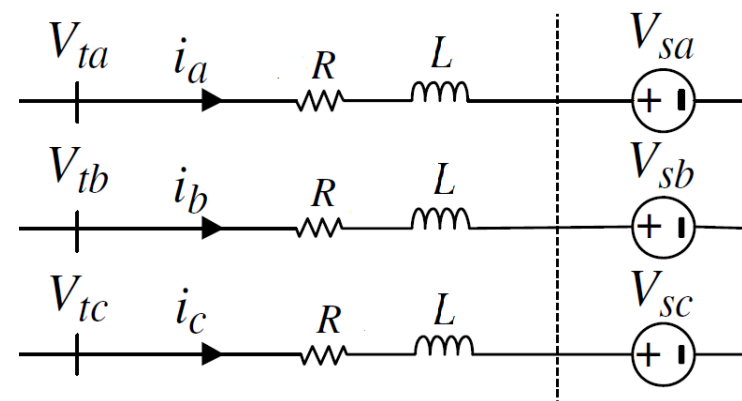
✓ i_d and i_q can be controlled by u_d and u_q respectively!

Grid-Following Converters

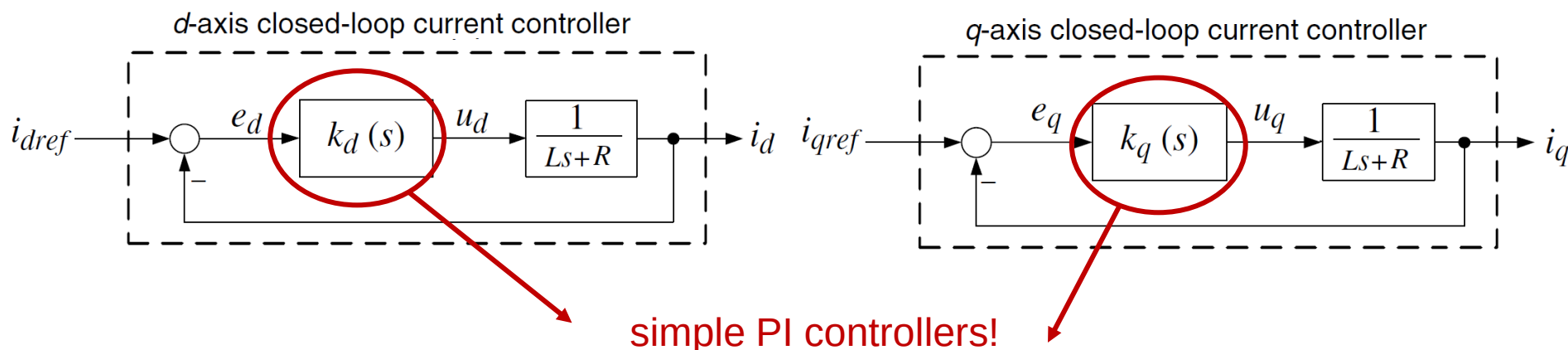
□ Dynamic Model of System in dq frame:

➤ Current controllers in dq frame:

$$\begin{cases} L \frac{di_d}{dt} = -Ri_d + u_d \\ L \frac{di_q}{dt} = -Ri_q + u_q \end{cases}$$



✓ Simplified current control loop based on u_d and u_q :



Grid-Following Converters

□ Dynamic Model of System in dq frame:

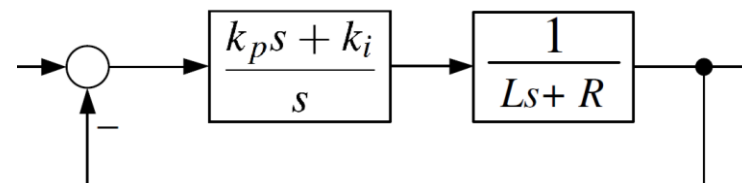
➤ Current controllers in dq frame:

✓ Let the current controllers be PI as:

$$k_q(s) = k_d(s) = k_p + \frac{k_i}{s}$$

loop gain
➔

$$\ell(s) = \left(\frac{k_p}{Ls} \right) \frac{s + k_i/k_p}{s + R/L}$$



✓ If **we want** to have the closed-loop **transfer function** be as:

$$\frac{I_d(s)}{I_{dref}(s)} = G_i(s) = \frac{\ell(s)}{1 + \ell(s)} = \frac{1}{\tau s + 1}$$



$$\begin{aligned} k_p &= L/\tau \\ k_i &= R/\tau \end{aligned}$$

selection of
time-constant
 τ

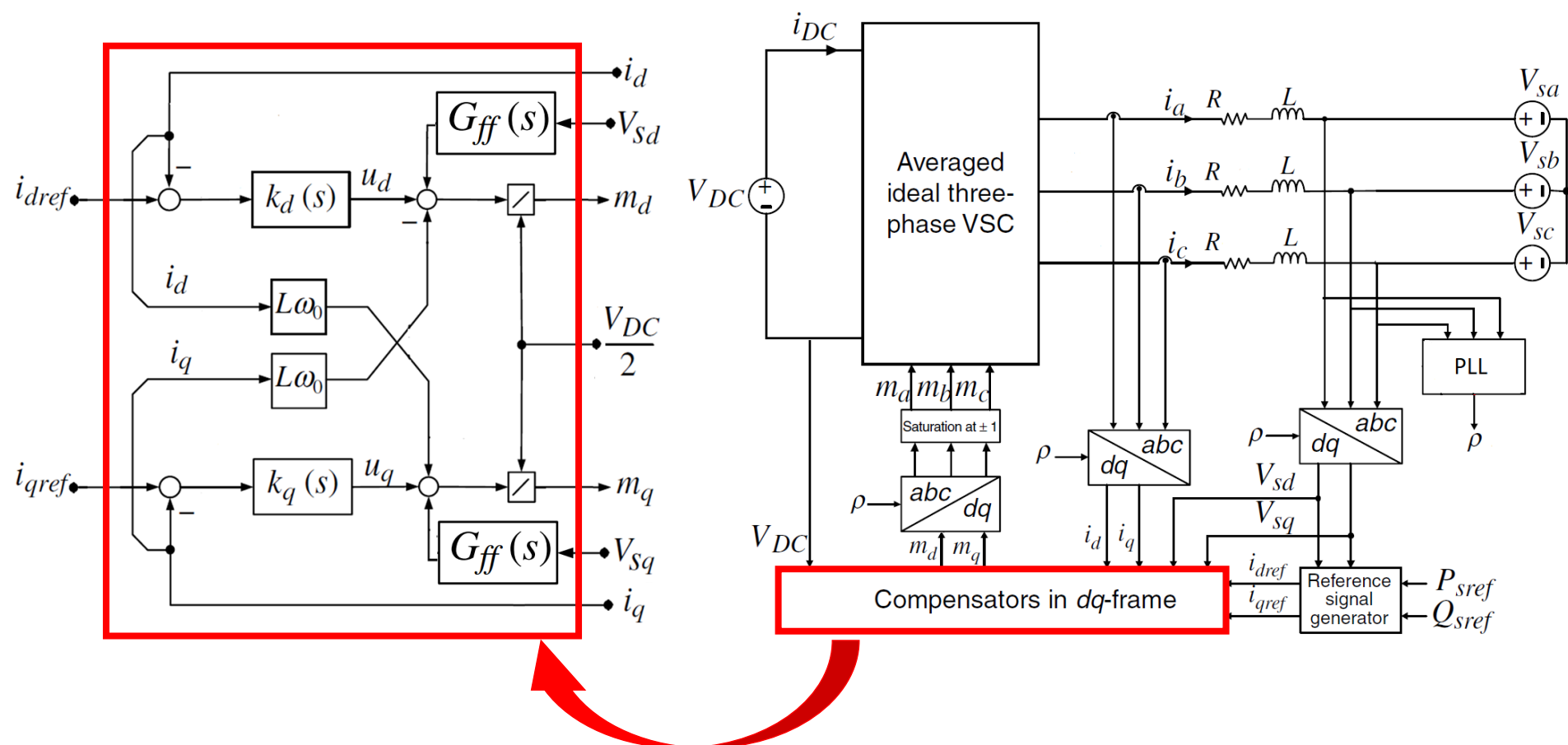
- ✓ small enough for fast current-control response!
- ✓ large enough so that the closed-loop bandwidth ($1/\tau$ in rad/s) is much smaller (~ 10 times) than the switching frequency of VSC!

Typical: $\tau \approx 0.5 - 5$ ms

Grid-Following Converters

□ Dynamic Model of System in dq frame:

❖ Complete Real/Reactive Power Controller in dq -frame:



Grid-Following Converters

□ Transient Study and with PQ Control:

➤ Example 8.2:

$$R = 1.63 \text{ m}\Omega \quad L = 100 \text{ }\mu\text{H}$$

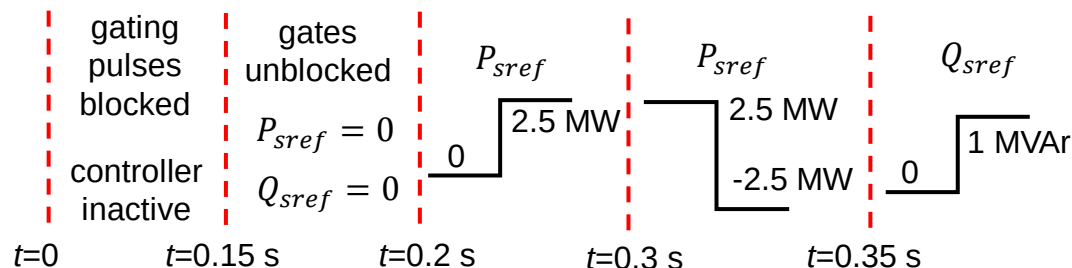
$$V_s^{LL} = 480 \text{ V} \quad \omega_0 = 377 \text{ rad/s}$$

$$V_{DC} = 1250 \text{ V} \quad f_{sw} = 3420 \text{ Hz}$$

$$G_{ff}(s) = 1/(8 \times 10^{-6}s + 1)$$

$$H_{PLL}(s) = \frac{685.42(s^2 + 568516)(s^2 + 166s + 6889)}{s(s^2 + 1508s + 568516)(s^2 + 964s + 232324)}$$

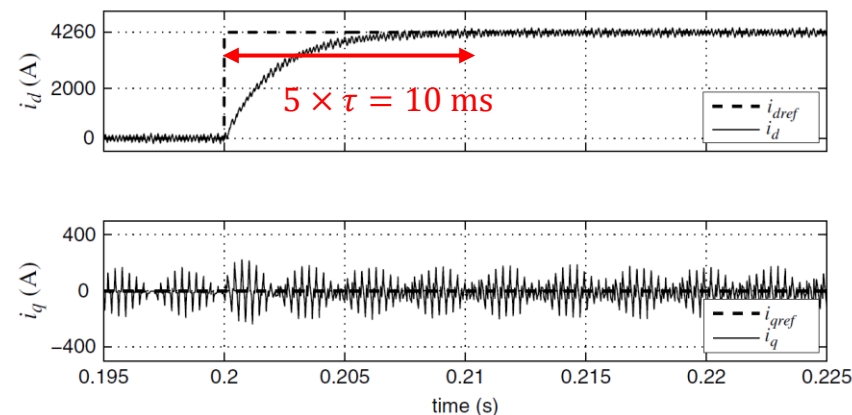
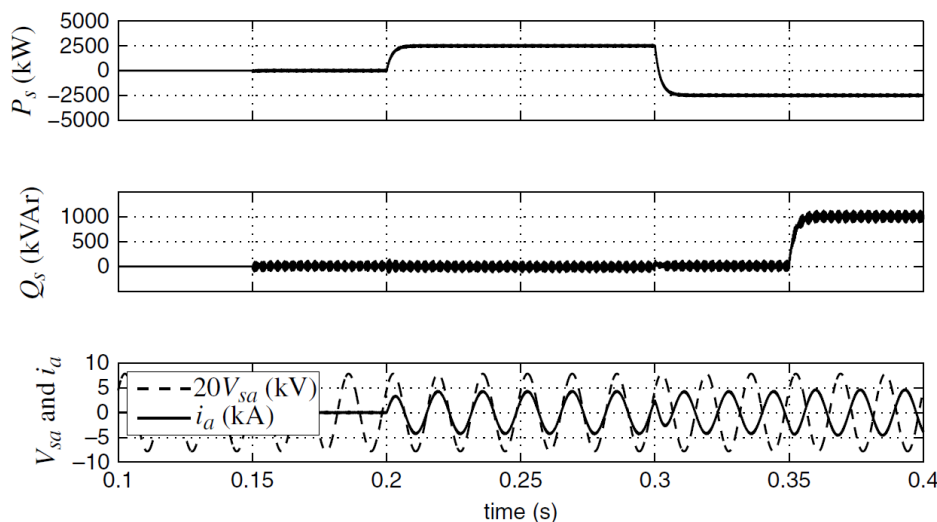
closed-loop time-constant $\tau = 2 \text{ ms}$



$$k_q(s) = k_d(s) = 0.05 + \frac{0.815}{s}$$



$$k_p = L/\tau = 0.05 \quad k_i = R/\tau = 0.815$$



Grid-Following Converters

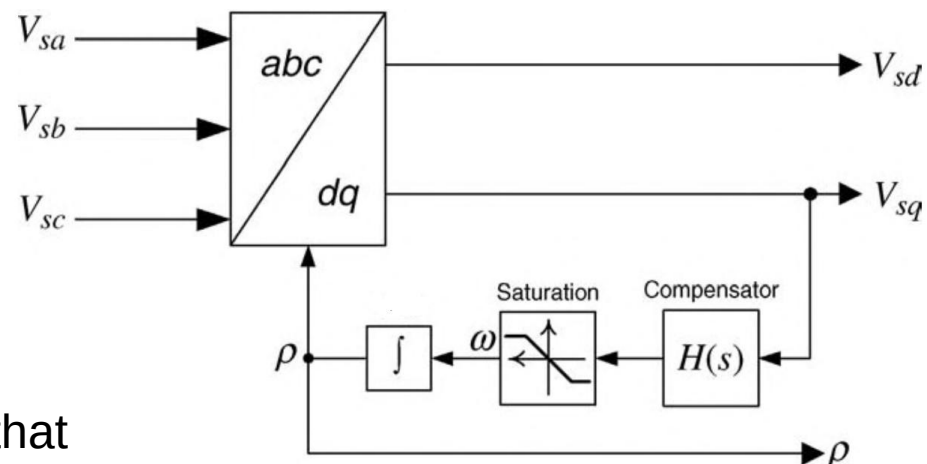
□ Synchronization with AC Grid:

❖ Phase-Locked Loop (PLL):

✓ a mechanism to ensure ($v_{sq} = 0$)

$$v_{sq} = \hat{V}_s \sin(\omega_0 t + \theta_0 - \rho) \quad \rho = \omega_0 t + \theta_0$$

➤ This is done using a compensator that adjusts speed to regulate v_{sq} at zero:



$$\omega(t) = H(p)v_{sq}(t)$$

$$\left\{ \begin{array}{l} \frac{d}{dt} \rho(t) = \omega(t) = H(p) \hat{V}_s \sin(\omega_0 t + \theta_0 - \rho) \\ \omega(0) = \omega_0 \\ \omega_{\min} < \omega < \omega_{\max} \end{array} \right. \quad \begin{array}{l} \approx \\ \approx 0 \end{array}$$

$$\frac{d\rho}{dt} = H(p) \hat{V}_s (\omega_0 t + \theta_0 - \rho)$$

✓ classical feedback control loop

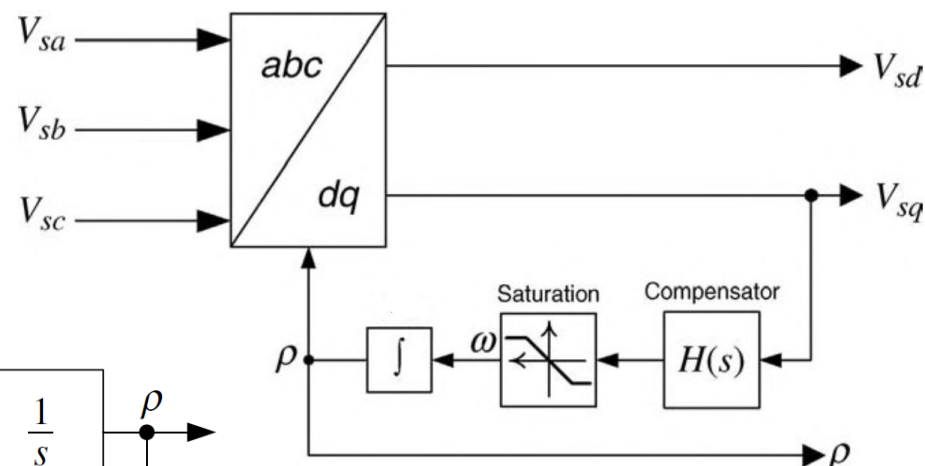
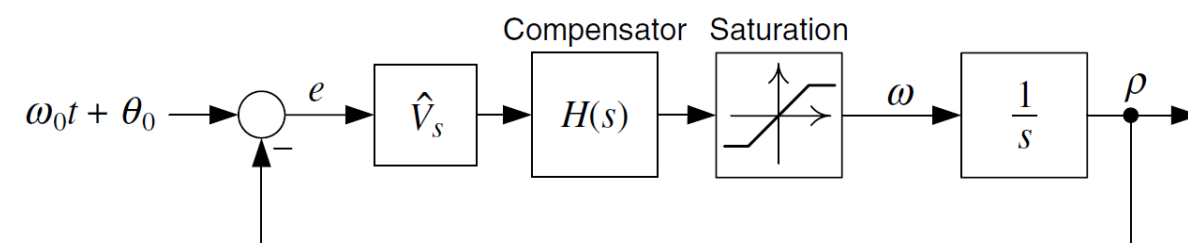
$$\begin{bmatrix} f_d(t) \\ f_q(t) \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos(\rho(t)) & \cos(\rho(t) - 2\pi/3) & \cos(\rho(t) + 2\pi/3) \\ -\sin(\rho(t)) & -\sin(\rho(t) - 2\pi/3) & -\sin(\rho(t) + 2\pi/3) \end{bmatrix} \begin{bmatrix} f_a(t) \\ f_b(t) \\ f_c(t) \end{bmatrix}$$

Grid-Following Converters

□ Synchronization with AC Grid:

❖ Phase-Locked Loop (PLL):

$$\frac{d\rho}{dt} = H(p)\hat{V}_s(\omega_0 t + \theta_0 - \rho)$$



❖ Compensator Design for PLL:

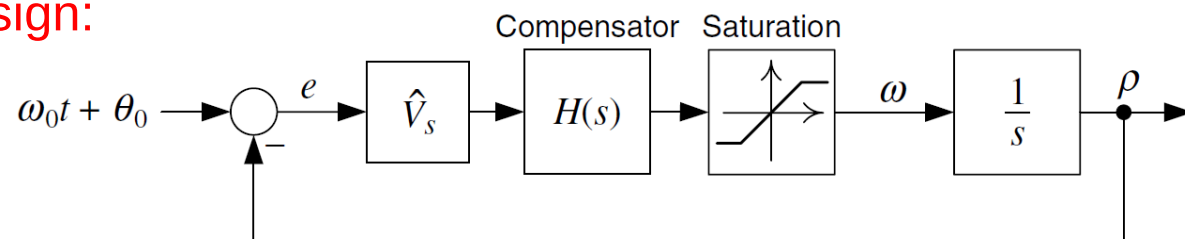
- ✓ ρ tracks the constant part of command (θ_0) with zero steady-state error because of the integral term in the loop
- ✓ To ensure zero steady-state error in tracking the ramp component ($\omega_0 t$) $H(s)$ must include at least one integrator [one pole at $s = 0$]
- ✓ Other poles and zeros of $H(s)$ are determined based on closed-loop bandwidth of PLL and stability indices [e.g. phase margin and gain margin]
- ✓ Unbalanced conditions/voltages should also be considered in design of compensator $H(s)$

Grid-Following Converters

□ Synchronization with AC Grid:

❖ PLL Compensator Design:

$$\frac{d\rho}{dt} = H(p)\hat{V}_s(\omega_0 t + \theta_0 - \rho)$$



➤ Assume unbalanced three-phase source voltages as:

$$\begin{cases} v_{sa}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0) + K_{neg} \hat{V}_s \cos(\omega_0 t + \theta_0) \\ v_{sb}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0 - 2\pi/3) + K_{neg} \hat{V}_s \cos(\omega_0 t + \theta_0 + 2\pi/3) \\ v_{sc}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0 + 2\pi/3) + K_{neg} \hat{V}_s \cos(\omega_0 t + \theta_0 - 2\pi/3) \end{cases}$$

$$\vec{v}_s(t) = \hat{V}_s e^{j(\omega_0 t + \theta_0)} + K_{neg} \hat{V}_s e^{-j(\omega_0 t + \theta_0)}$$

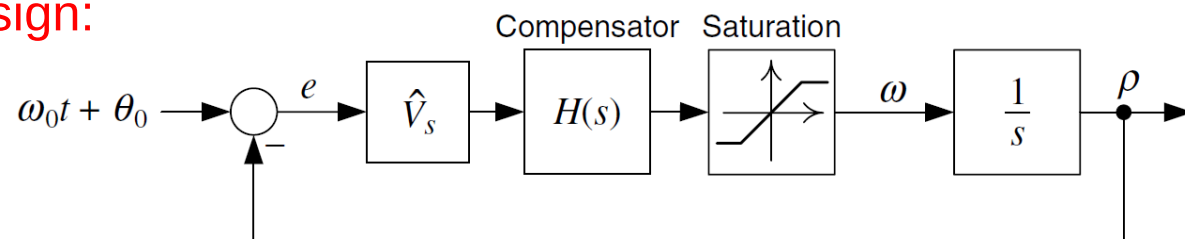
PLL in steady-state: $\rho = \omega_0 t + \theta_0$ ➡ $v_{sdq} = \vec{v}_s e^{-j\rho} = \hat{V}_s + K_{neg} \hat{V}_s e^{-j(2\omega_0 t + 2\theta_0)}$

Grid-Following Converters

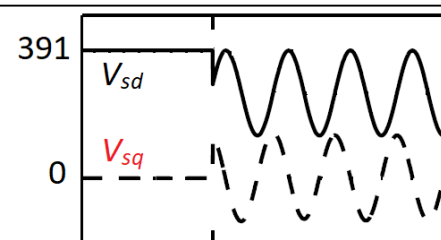
□ Synchronization with AC Grid:

❖ PLL Compensator Design:

$$\frac{d\rho}{dt} = H(p)\hat{V}_s(\omega_0 t + \theta_0 - \rho)$$



$$\begin{cases} v_{sd} = \hat{V}_s + K_{neg} \hat{V}_s \cos(2\omega_0 t + 2\theta_0) \\ v_{sq}(t) = 0 - K_{neg} \hat{V}_s \sin(2\omega_0 t + 2\theta_0) \end{cases}$$



typical: $k_{neg} \approx 0.01$ ➡ $k_{neg} \approx 0.5$ ➡ attenuating the double-frequency ($2\omega_0$) sinusoidal component in V_{sq} is essential

single-phase to ground faults

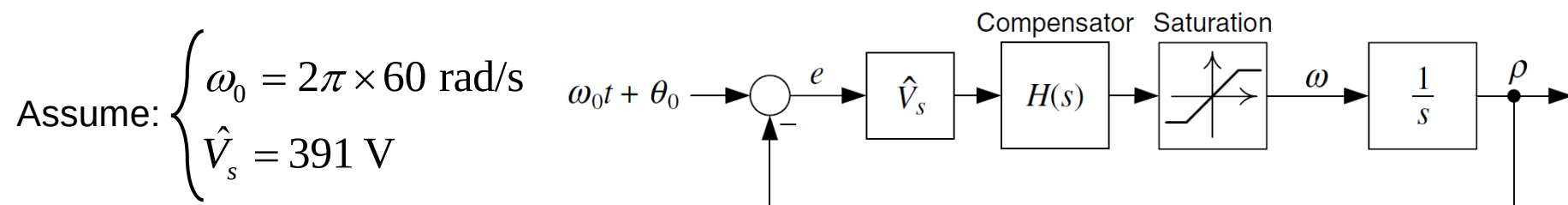
attenuation of ($2\omega_0$) component by $H(s)$ {

- Approach #1:** ensure $H(s)$ exhibits strong low-pass characteristic!
may compromise the PLL closed-loop bandwidth!
- Approach #2:** include in $H(s)$ one pair of complex-conjugate zeros at $(s = \pm j2\omega_0)$
eliminates the double-frequency ripple in v_{sq} without sacrificing the PLL closed-loop bandwidth and it can be selected large!

Grid-Following Converters

□ Synchronization with AC Grid:

❖ PLL Compensator Design – Example 8.1:



Need: Gain crossover frequency $\omega_c = 200 \text{ rad/s}$ and phase margin of 60 degrees

$H(s)$ must include:

- 1) one pole at $(s = 0)$
- 2) complex-conjugate zeros at $(s = \pm j2\omega_0)$
- 3) double real pole at $(s = -2\omega_0)$

➔ To ensure the loop gain magnitude continues to drop with -40 dB/dec for $\omega > 2\omega_0$

➔ $H(s) = \left(\frac{1}{\hat{V}_s} \right) \frac{s^2 + (2\omega_0)^2}{s(s + 2\omega_0)^2} F(s)$ ➔ loop gain $\ell(s) = \frac{s^2 + (2\omega_0)^2}{s^2 (s + 2\omega_0)^2} F(s)$

➔ $\begin{cases} |\ell(s = j200)| = 1 \\ \angle \ell(s = j200) = \text{PM} - 180^\circ \end{cases} \quad \Rightarrow \quad \begin{cases} |F(s = j200)| = \frac{1}{2.1713 \times 10^{-5}} = 46055 \\ \angle F(s = j200) = -120^\circ - (-210^\circ) = 90^\circ \end{cases}$

Grid-Following Converters

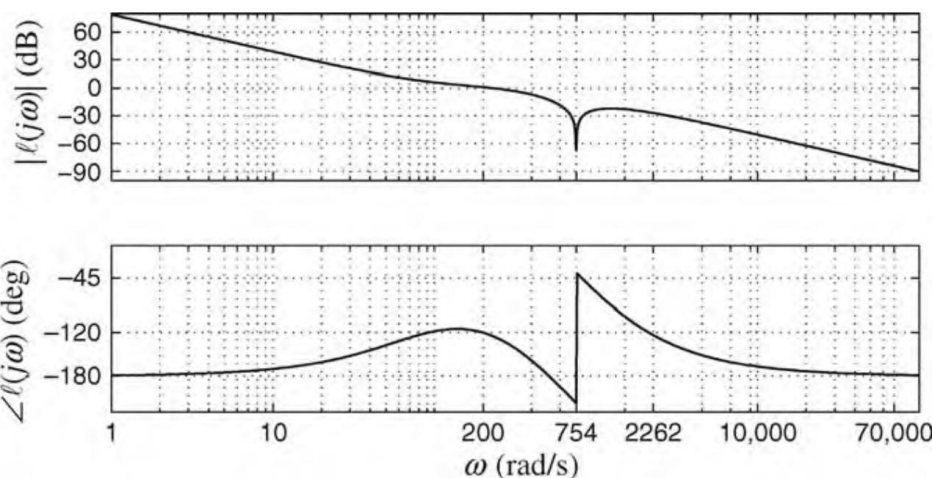
□ Synchronization with AC Grid:

$$\ell(s) = \frac{s^2 + (2\omega_0)^2}{s^2 (s + 2\omega_0)^2} F(s) \quad \Rightarrow \quad \begin{cases} |F(s = j200)| = \frac{1}{2.1713 \times 10^{-5}} = 46055 \\ \angle F(s = j200) = -120^\circ - (-210^\circ) = 90^\circ \end{cases}$$

✓ Use two cascaded lead compensators to get the 90° phase advance, 45° each at 200 rad/s!

$$F(s) = h \frac{(s + p/\alpha)(s + p/\alpha)}{(s + p)(s + p)} \quad \Rightarrow \quad \begin{cases} \alpha = \frac{1 + \sin(\delta_m)}{1 - \sin(\delta_m)} \\ p = \omega_c \sqrt{\alpha} \end{cases} \quad \begin{matrix} \delta_m = 45^\circ \\ \omega_c = 200 \end{matrix} \quad \begin{cases} \alpha = 5.8 \\ p = 482 \end{cases}$$

$$\Rightarrow F(s) = h \frac{(s + 83)^2}{(s + 482)^2} \quad \begin{matrix} |F(s = j200)| = 46055 \\ h = \frac{46055}{0.1722} = 2.68 \times 10^5 \end{matrix}$$



$$\ell(s) = h \times \frac{s^2 + (2\omega_0)^2}{s^2 (s + 2\omega_0)^2} \times \frac{(s + 83)^2}{(s + 482)^2} \quad \omega_0 = 377 \text{ rad/s}$$

$$\ell(s) = 2.68 \times 10^5 \times \frac{s^2 + 754^2}{s^2 (s + 754)^2} \times \frac{(s + 83)^2}{(s + 482)^2} \quad \hat{V}_s = 391 \text{ V}$$

$$H(s) = \frac{\ell(s)}{\hat{V}_s} = 685.42 \times \frac{s^2 + 754^2}{s^2 (s + 754)^2} \times \frac{(s + 83)^2}{(s + 482)^2}$$

Grid-Following Converters

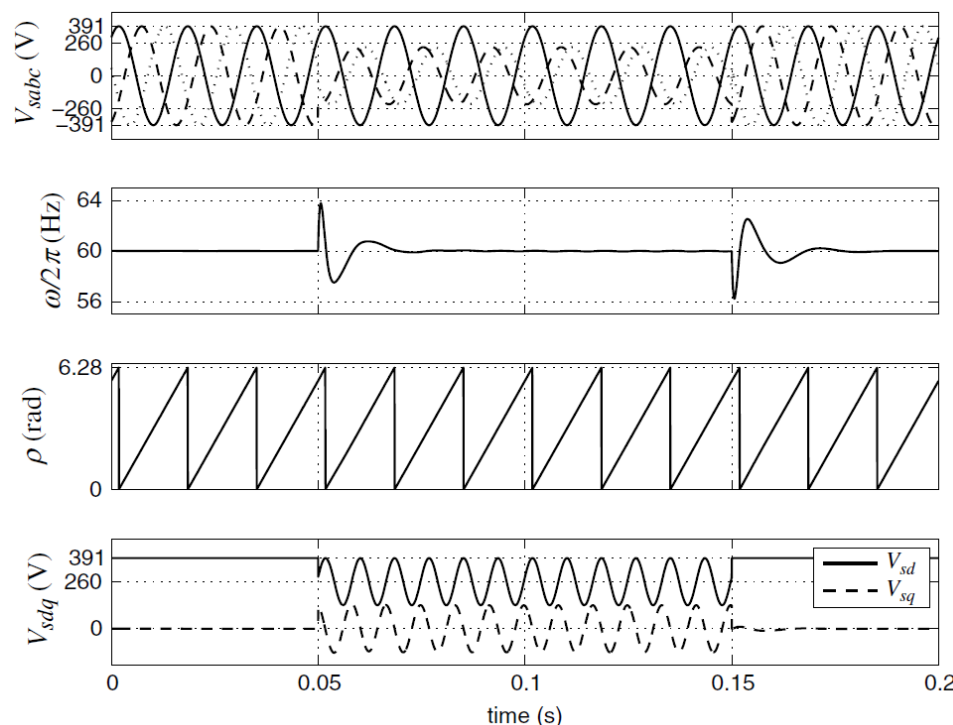
□ Synchronization with AC Grid:

❖ PLL Compensator Design – Example 8.1:

Transient Study – Sudden imbalance in three-phase source voltages:

At $t=0.05$ s: $\hat{V}_s$ drops from 391 V to 260 V, and k_{neg} steps from 0 to 0.5

At $t=0.15$ s: back to pre-fault



✓ v_{sd} and v_{sq} include 120Hz sinusoidal ripple during fault due to the negative sequence voltages

✓ compensator tracks 60Hz while changing ω to maintain the average value of v_{sq} at zero

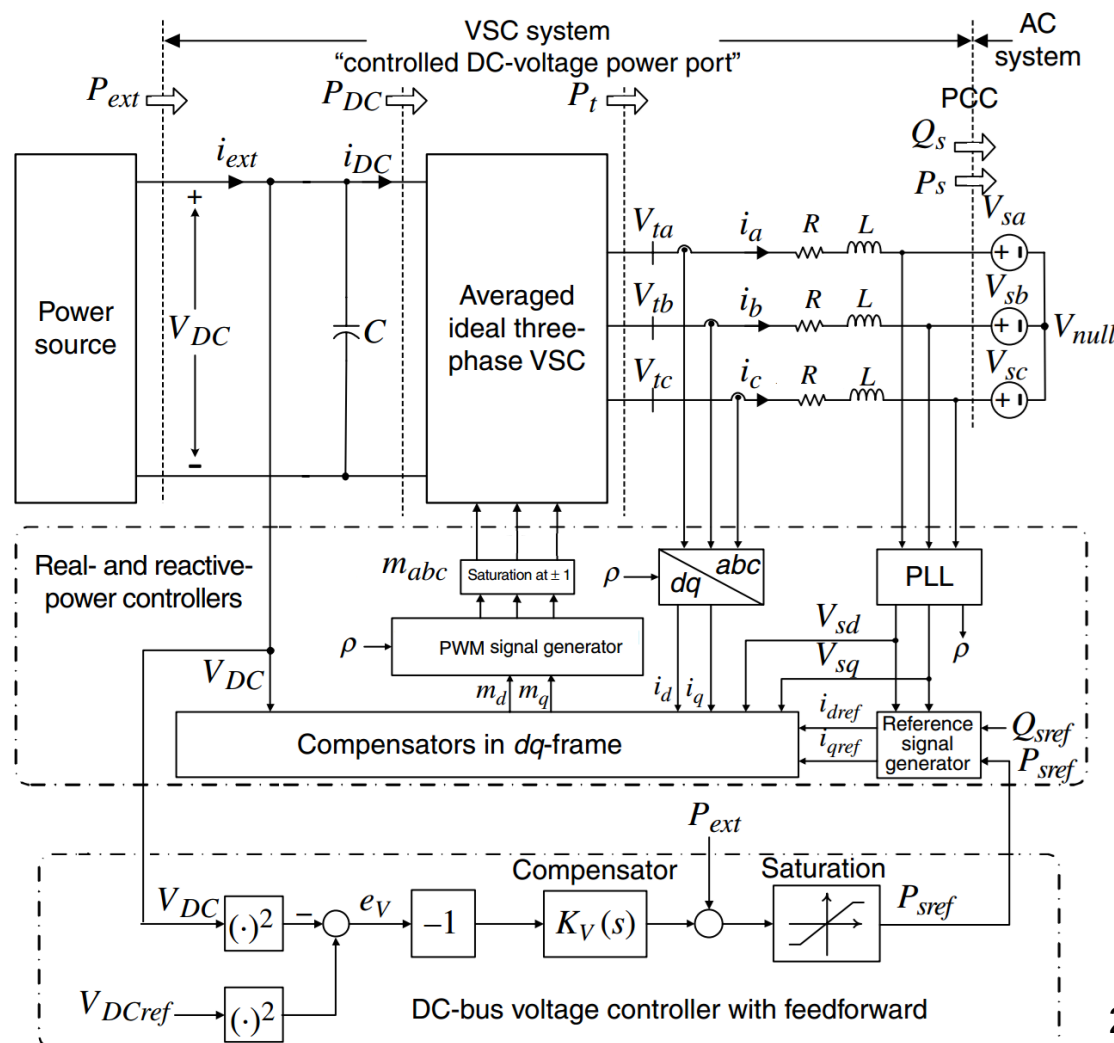
✓ The ripple is suppressed by $H(s)$ where ω and ρ remain free of distortion

Grid-Following Converters

□ Controlled DC-Voltage Power Port:

❖ Objective: Regulate the DC Bus Voltage!

- The kernel is still the real-reactive power controller!
- ✓ A feedback mechanism adjusts P_s to regulate V_{DC} through a compensator!
- This is done by regulating the energy stored in the capacitor of the dc-link!
- ✓ Reactive power Q_s is controlled independently, (typically at Zero!)
- ✓ Q_s can also be controlled to regulate the PCC voltage (like in STATCOM)!



Grid-Following Converters

□ Controlled DC-Voltage Power Port:

❖ **Objective:** Let's regulate V_{DC}^2 instead of V_{DC} !

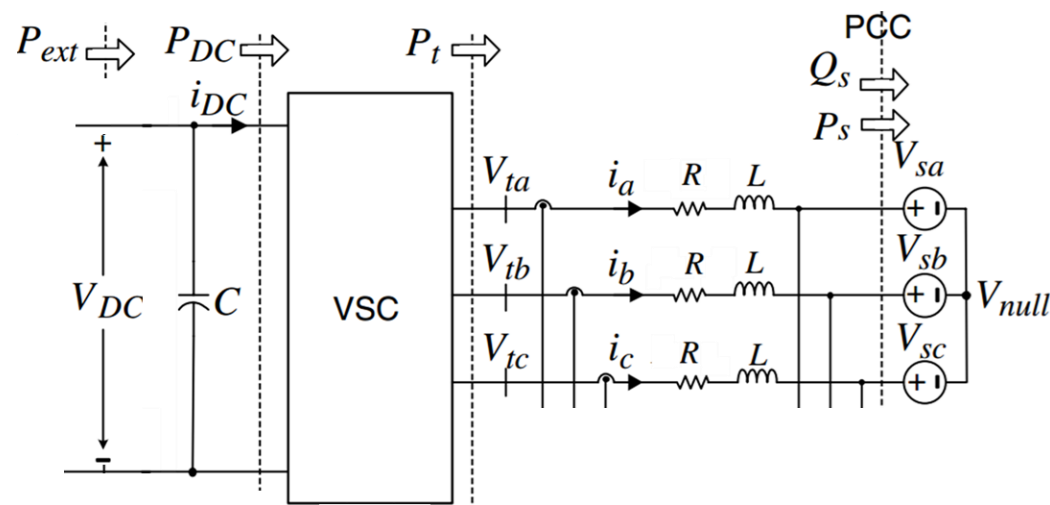
❖ Dynamics of V_{DC}^2 :

Power Balance:

$$\frac{1}{2}C \frac{dV_{DC}^2}{dt} = P_{ext} - P_t$$

→ $P_t = P_s + P_L$

→ $P_L = i_a L \frac{di_a}{dt} + i_b L \frac{di_b}{dt} + i_c L \frac{di_c}{dt}$



$$\begin{bmatrix} f_a \\ f_b \\ f_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \\ -1/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} f_\alpha \\ f_\beta \end{bmatrix}$$

$$P_s = \frac{3}{2} [v_{s\alpha} i_\alpha + v_{s\beta} i_\beta]$$

$$Q_s = \frac{3}{2} [v_{s\beta} i_\alpha - v_{s\alpha} i_\beta]$$

Grid-Following Converters

Controlled DC-Voltage Power Port:

❖ **Objective:** Let's regulate V_{DC}^2 instead of V_{DC} !

❖ **Dynamics of V_{DC}^2 :**

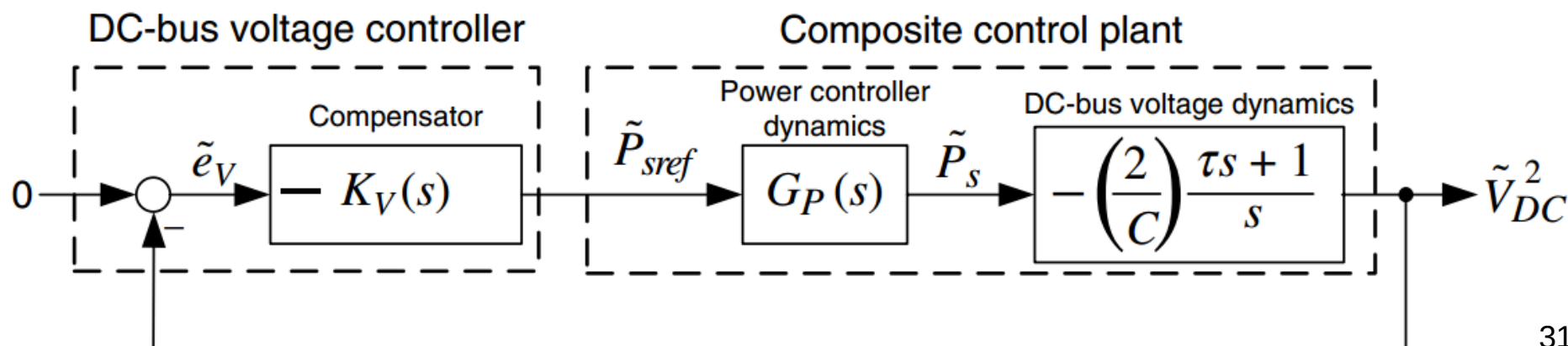
$$\frac{dV_{DC}^2}{dt} = \frac{2}{C} \left\{ P_{ext} - P_s - \left(\frac{2L}{3\hat{V}_s^2} \right) \left(P_s \frac{dP_s}{dt} + Q_s \frac{dQ_s}{dt} \right) \right\}$$

The control plant is nonlinear due to $P_s \frac{dP_s}{dt}$ linearize

$$G_v(s) = \frac{\tilde{V}_{DC}^2(s)}{\tilde{P}_s(s)} = - \left(\frac{2}{C} \right) \frac{1 + \tau s}{s}$$

Transfer function in Laplace domain from $\tilde{P}_s$ to $\tilde{V}_{DC}^2$

where $\tau = \frac{2LP_{s0}}{3\hat{V}_s^2} = \frac{2LP_{ext0}}{3\hat{V}_s^2}$



Grid-Following Converters

Controlled DC-Voltage Power Port:

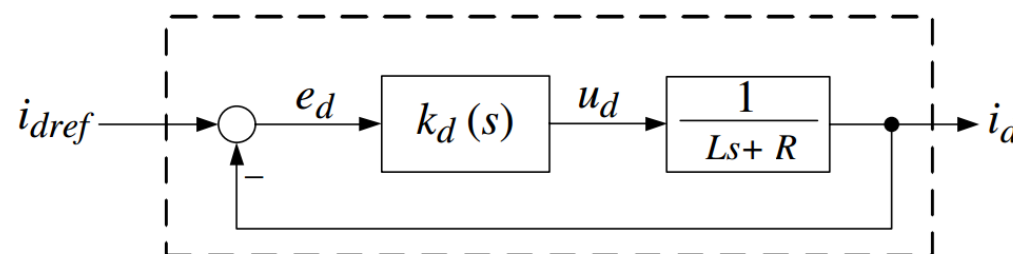
❖ **Objective:** Let's regulate V_{DC}^2 instead of V_{DC} !

$$G_v(s) = \frac{\tilde{V}_{DC}^2(s)}{\tilde{P}_s(s)} = -\left(\frac{2}{C}\right) \frac{1 + \tau_v s}{s} \quad \text{where} \quad \tau_v = \frac{2LP_{s0}}{3V_{sd}^2} = \frac{2LP_{ext0}}{3V_{sd}^2}$$

➤ **Recall:**

$$\frac{I_d(s)}{I_{dref}(s)} = G_i(s) = \frac{1}{\tau_i s + 1}$$

$$I_d(s) = G_i(s) I_{dref}(s)$$



$$k_d(s) = k_p + \frac{k_i}{s}$$

$$k_p = L/\tau_i \quad k_i = R/\tau_i$$

$$P_s(s) = G_i(s) P_{sref}(s)$$

$$P_s = \frac{3}{2} v_{sd} i_d$$

multiply by

 V_{sd}



$$G_P(s) = \frac{P_s(s)}{P_{sref}(s)} = G_i(s) = \frac{1}{\tau_i s + 1}$$

Grid-Following Converters

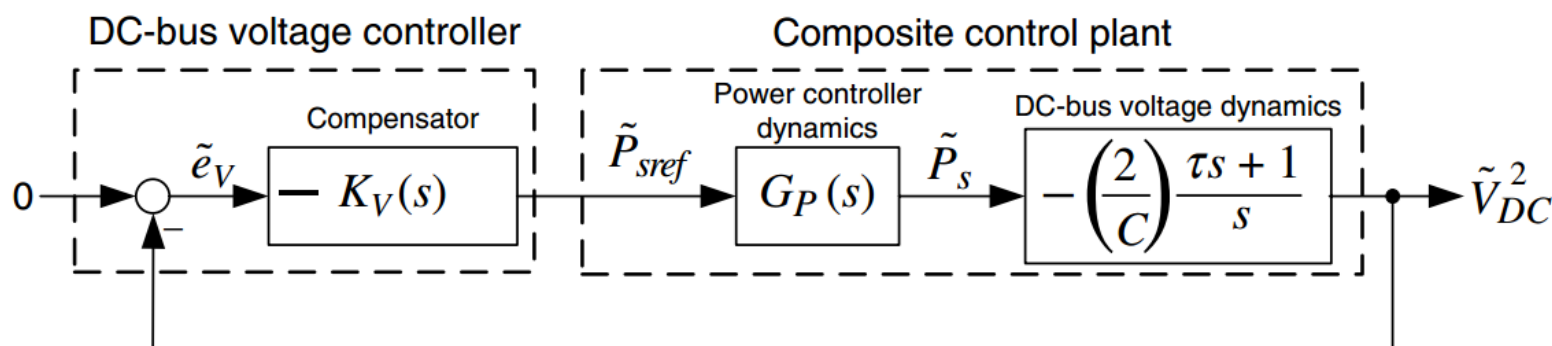
□ Controlled DC-Voltage Power Port:

❖ Compensator Design for DC Bus Voltage Controller:

- ✓ To eliminate the steady-state error due to losses, etc., $Kv(s)$ must include one integrator
- ✓ To ensure adequate phase margin and stability, $Kv(s)$ must also include one zero

➤ Let : $K_v(s) = \frac{1}{s} \times H(s) \Rightarrow \ell(s) = -K_v(s)G_p(s)G_v(s) = \frac{2}{C} H(s) \frac{(1 + \tau_v s)}{s^2 (1 + \tau_i s)}$

- ✓ Select $H(s)$ such that $\ell(j\omega)$ crossed 0dB axis with a slope of about -20 dB/dec at $\omega = \omega_c$ and phase of $\ell(j\omega_c)$ be larger than -180° by a reasonable phase margin
- ✓ Note that τ_v depends on the real power $\Rightarrow$ ensure adequate phase margin at rated power (worst-case)
- ✓ Choose ω_c 2-10 times lower than the bandwidth of G_p so that G_p does not introduce much phase delay and reduces phase margin (effectively $G_p(j\omega_c) \approx 1 + j0$)



Grid-Following Converters

□ Controlled DC-Voltage Power Port:

❖ Compensator Design for DC Bus Voltage Controller— Example 8.4:

Assume: $\begin{cases} R = 3.26 \text{ m}\Omega & L = 200 \text{ }\mu\text{H} & C = 9625 \text{ }\mu\text{F} \\ V_{sd} = 391 \text{ V} & \omega_0 = 377 \text{ rad/s} & P_s = \pm 2.5 \text{ MW} \\ V_{DC}^{\text{rated}} = 2500 \text{ V} & f_{sw} = 1680 \text{ Hz} & \tau_i = 1.0 \text{ ms} \end{cases}$

$\Rightarrow \begin{cases} k_p = L/\tau_i = 0.2 \text{ }\Omega \\ k_i = R/\tau_i = 3.26 \text{ }\Omega/\text{s} \\ G_p(s) = \frac{1}{\tau_i s + 1} = \frac{1}{0.001s + 1} \end{cases}$

Note 1: The worst-case in terms of phase of $G_v(s)$ corresponds to negative rated power

$\Rightarrow \tau_v = \frac{2LP_{s0}}{3V_{sd}^2} = -0.0022$

Note 2: The bandwidth of $G_p(s)$ is 1000 rad/s $\Rightarrow$ choose 1/5 of it for $\omega_c = 200 \text{ rad/s}$

$\triangleright$ Let: $K_v(s) = \frac{1}{s} \times H(s) \Rightarrow \ell(s) = H(s) \frac{2}{9625 \times 10^{-6}} \frac{(1 - 0.0022s)}{s^2 (1 + 0.001s)} = H(s) \times N(s)$

$\Rightarrow \begin{cases} |N(s = j200)| = 0.0056 \\ \angle N(s = j200) = -215^\circ \end{cases} \Rightarrow \begin{cases} |H(s = j200)| = 1/0.0056 = 178.6 \\ \angle H(s = j200) = -135^\circ - (-215^\circ) = 80^\circ \end{cases}$

$\begin{cases} |\ell(s = j200)| = 1 \\ \angle \ell(s = j200) = \text{PM} - 180^\circ = -135^\circ \end{cases}$

desired: loop gain $\Rightarrow$ Use a lead compensator $H(s) = h \frac{(s + p/\alpha)}{(s + p)}$

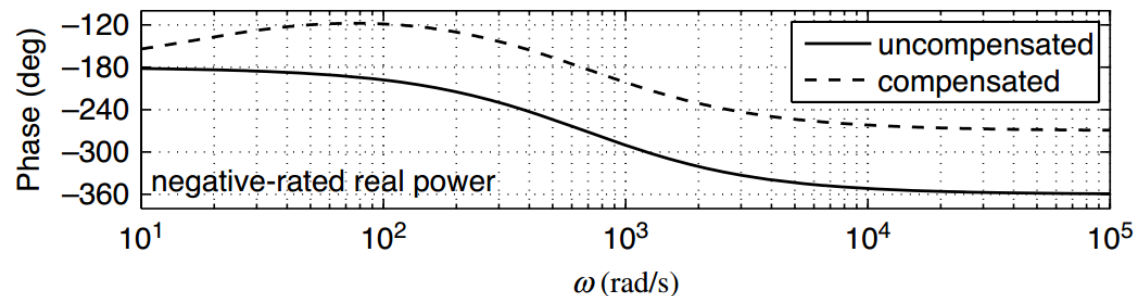
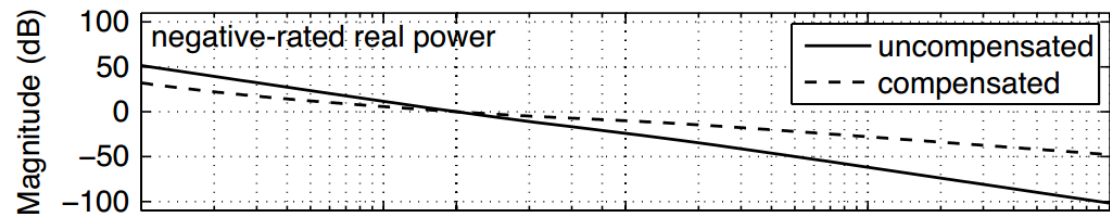
Grid-Following Converters

Controlled DC-Voltage Power Port:

Compensator Design for DC Bus Voltage Controller— Example 8.4:

$$H(s) = h \frac{(s + p/\alpha)}{(s + p)} \rightarrow \begin{cases} \alpha = \frac{1 + \sin(\delta_m)}{1 - \sin(\delta_m)} & \delta_m = 80^\circ \\ p = \omega_c \sqrt{\alpha} & \omega_c = 200 \end{cases} \rightarrow \begin{cases} \alpha = 130.65 \\ p = 2286 \end{cases}$$

$$\rightarrow H(s) = h \frac{(s + 17.5)}{(s + 2286)} \rightarrow |H(s = j200)| = 178.6 \quad h = \frac{178.6}{0.0875} = 2041$$



$$K_v(s) = 2041 \times \frac{(s + 17.5)}{s(s + 2286)}$$

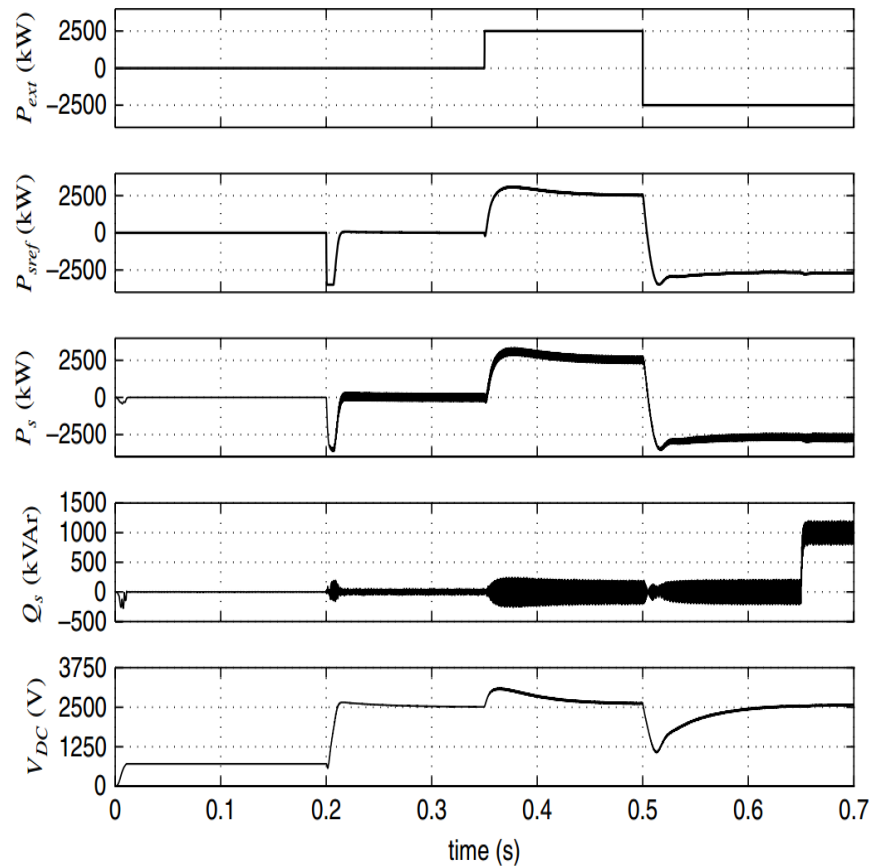
$$\ell(s) = 4.24 \times 10^5 \times \frac{(s + 17.5)}{s(s + 2286)} \frac{(1 - 0.0022s)}{s^2(1 + 0.001s)}$$

Grid-Following Converters

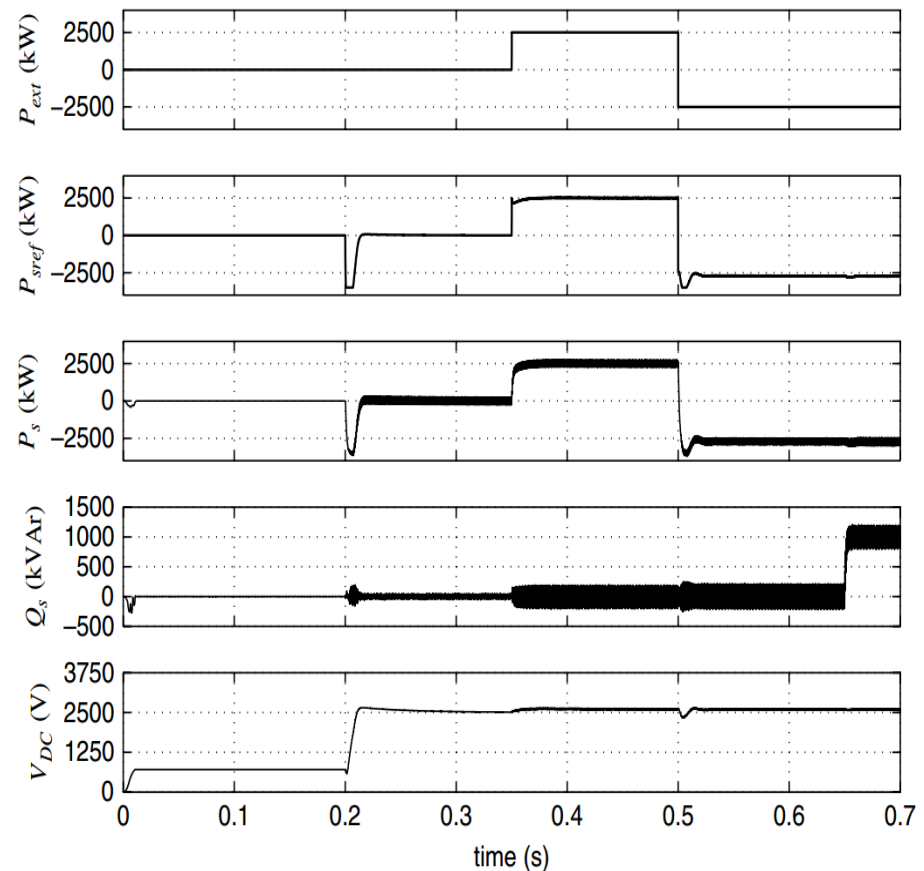
□ Controlled DC-Voltage Power Port:

❖ Compensator Design for DC Bus Voltage Controller— Example 8.4:

Without Feedforward



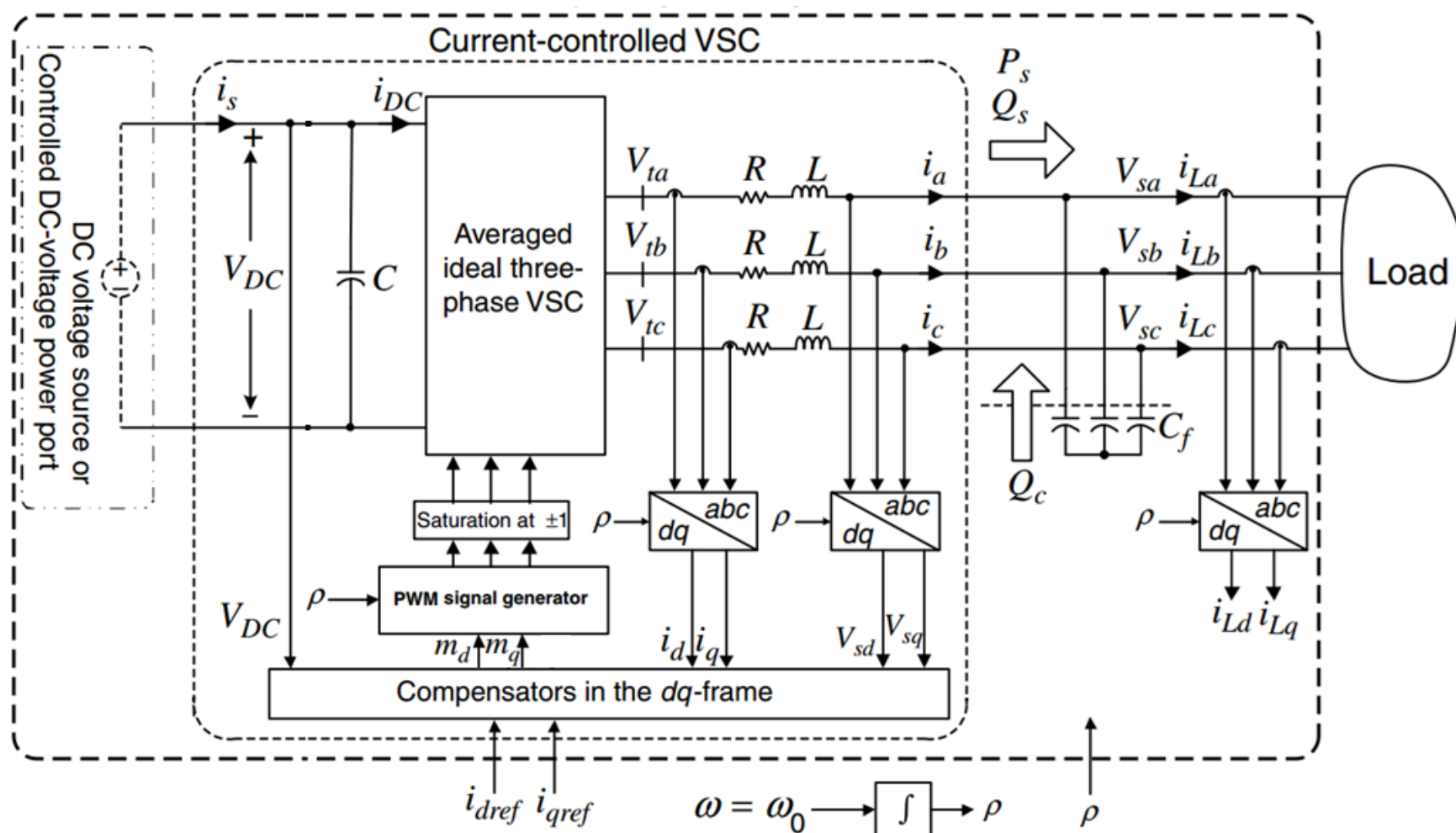
With Feedforward



Grid-Forming Converters

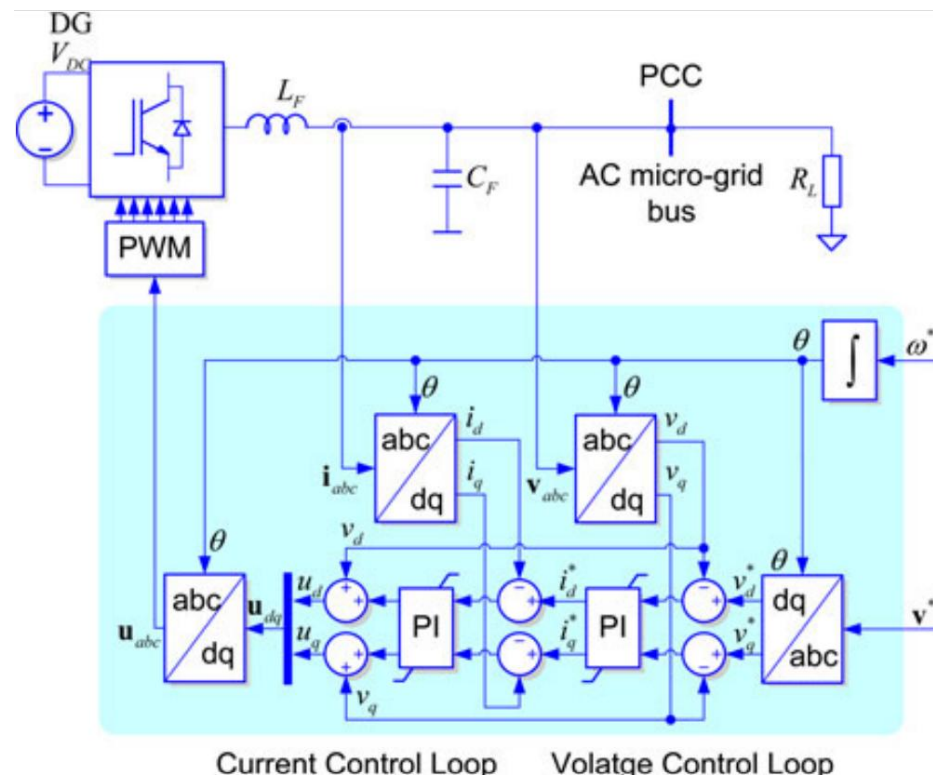
□ Form a Grid/Source to Supply the Load:

❖ Objective: Control voltage and frequency at PCC!



Control of Power Converters in AC Microgrids

IEEE TRANSACTIONS ON POWER ELECTRONICS, VOL. 27, NO. 11, NOVEMBER 2012



Grid Forming (GFM)

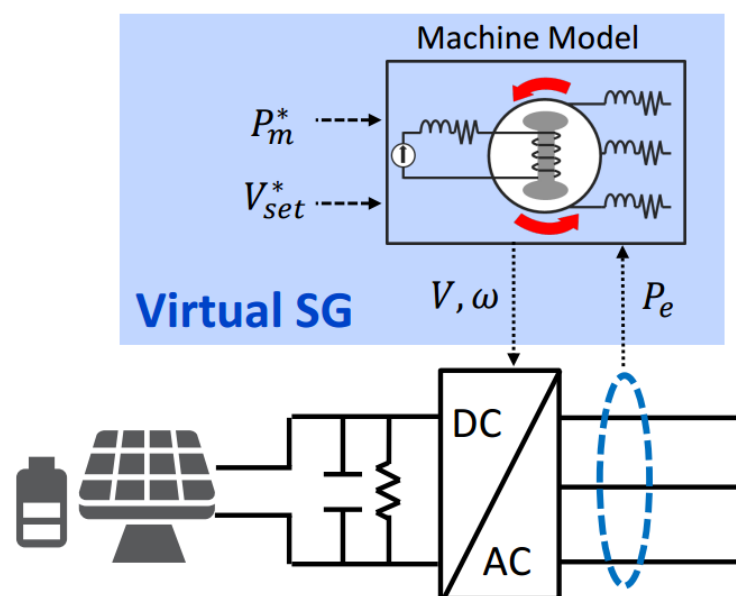
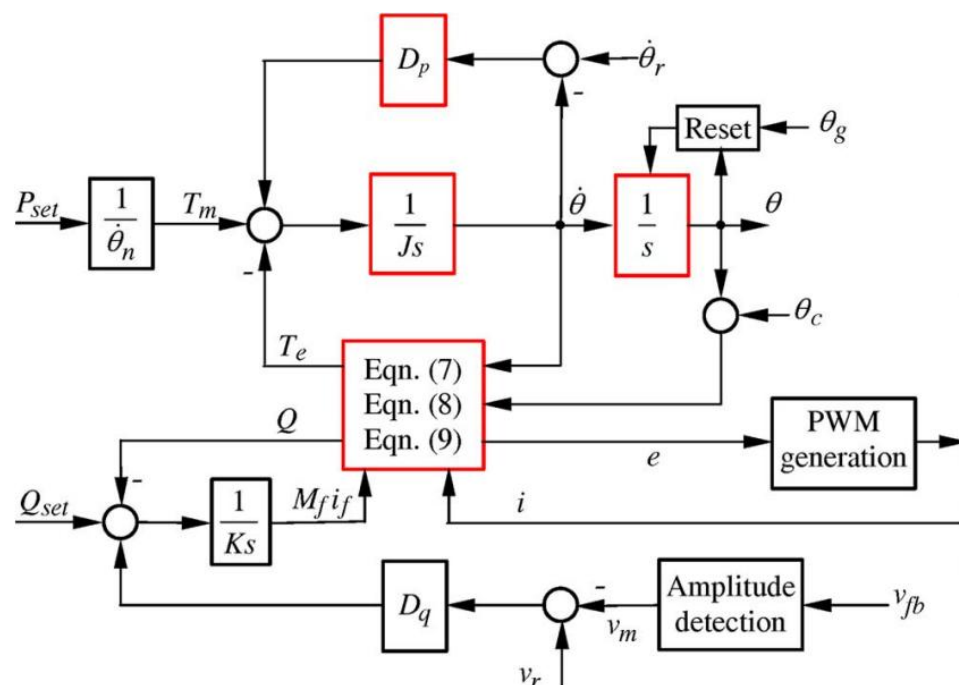
Grid-Forming Converters

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS, VOL. 58, NO. 4, APRIL 2011

Synchronverters: Inverters That Mimic Synchronous Generators

Qing-Chang Zhong, *Senior Member, IEEE*, and George Weiss

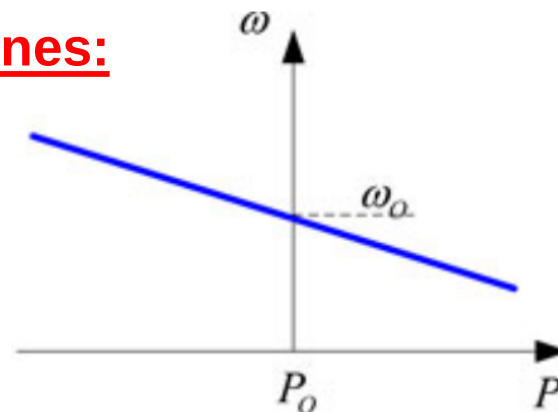
❖ Virtual Synchronous Generator (VSG):



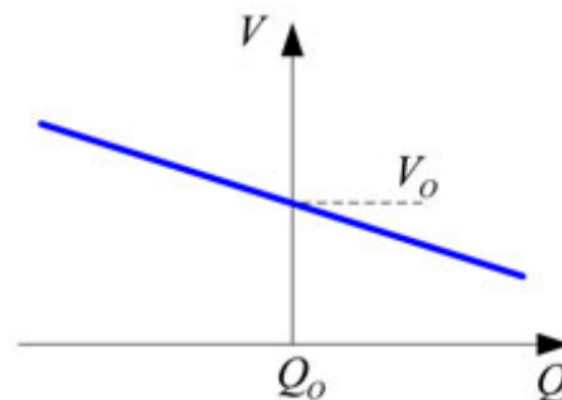
Grid-Forming Converters

❖ Droop Control:

➤ Inductive Lines:

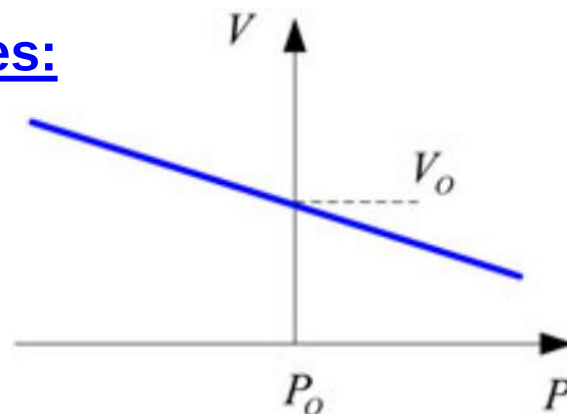


$$\omega = \omega_0 - k_p (P - P_0)$$

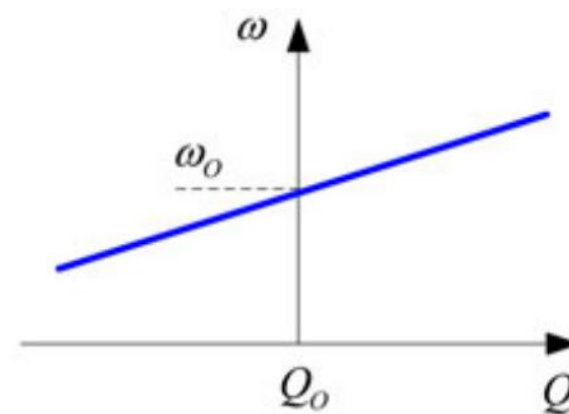


$$V = V_0 - k_q (Q - Q_0)$$

➤ Resistive Lines:



$$V = V_0 - k_p (P - P_0)$$

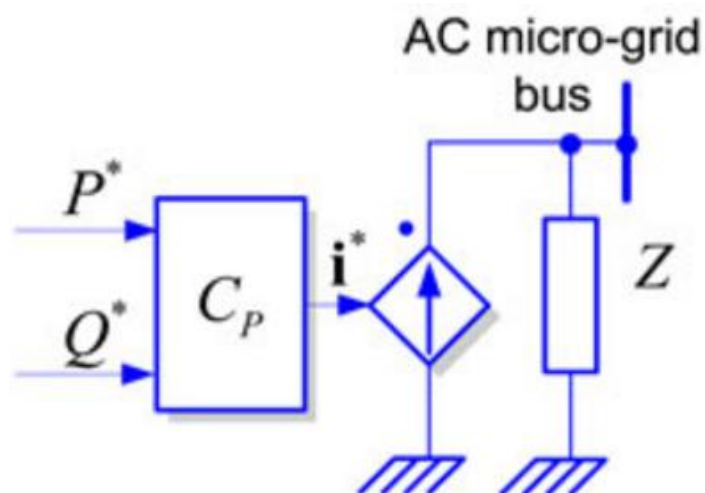


$$\omega = \omega_0 + k_q (Q - Q_0)$$

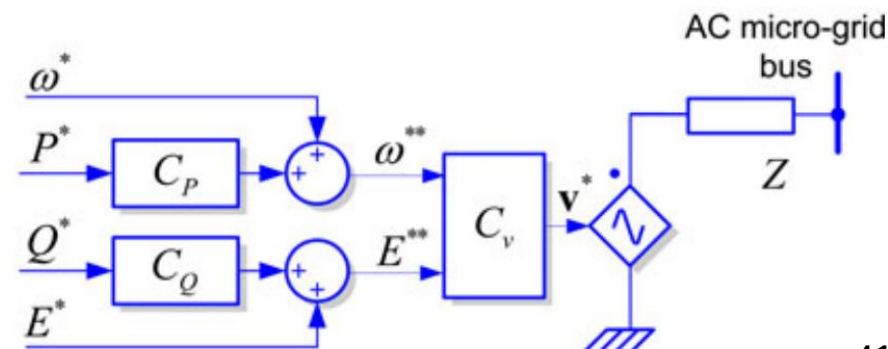
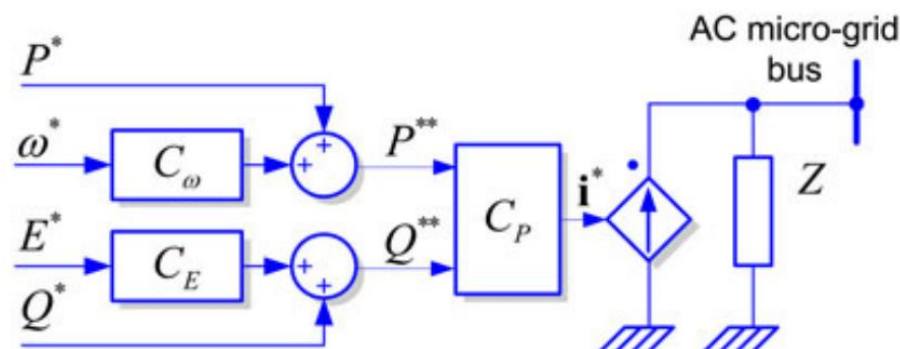
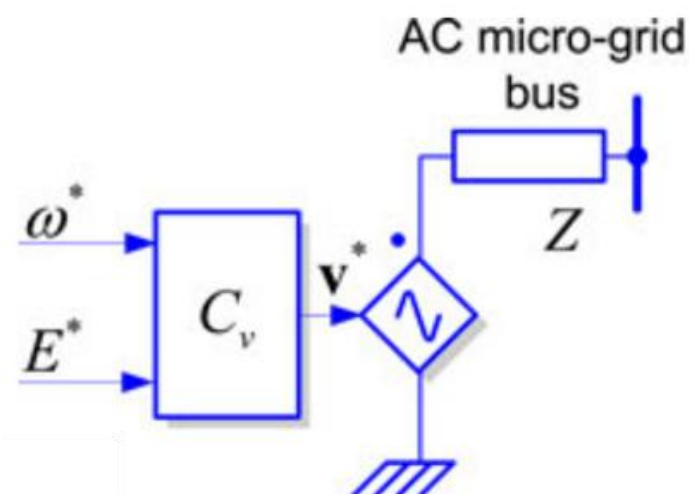
Grid-Forming Converters

❖ Representation of Power Converters:

Grid Following (GFL)



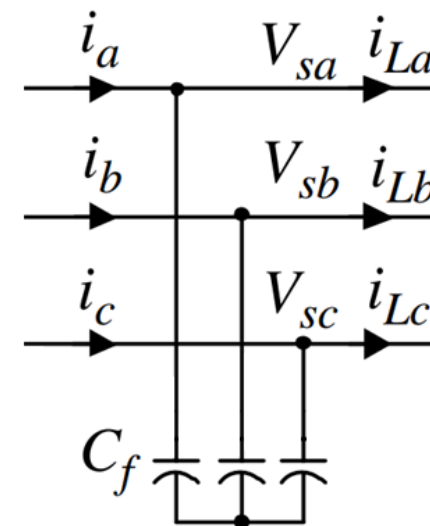
Grid Forming (GFM)



Grid-Forming Converters

□ Modeling of the System:

$$C_f \frac{d\vec{v}_s}{dt} = \vec{i} - \vec{i}_L \quad \Rightarrow \quad C_f \frac{d}{dt} \left[\left(v_{sd} + jv_{sq} \right) e^{j\rho} \right] = \left(i_d + ji_q \right) e^{j\rho} - \left(i_{Ld} + ji_{Lq} \right) e^{j\rho}$$



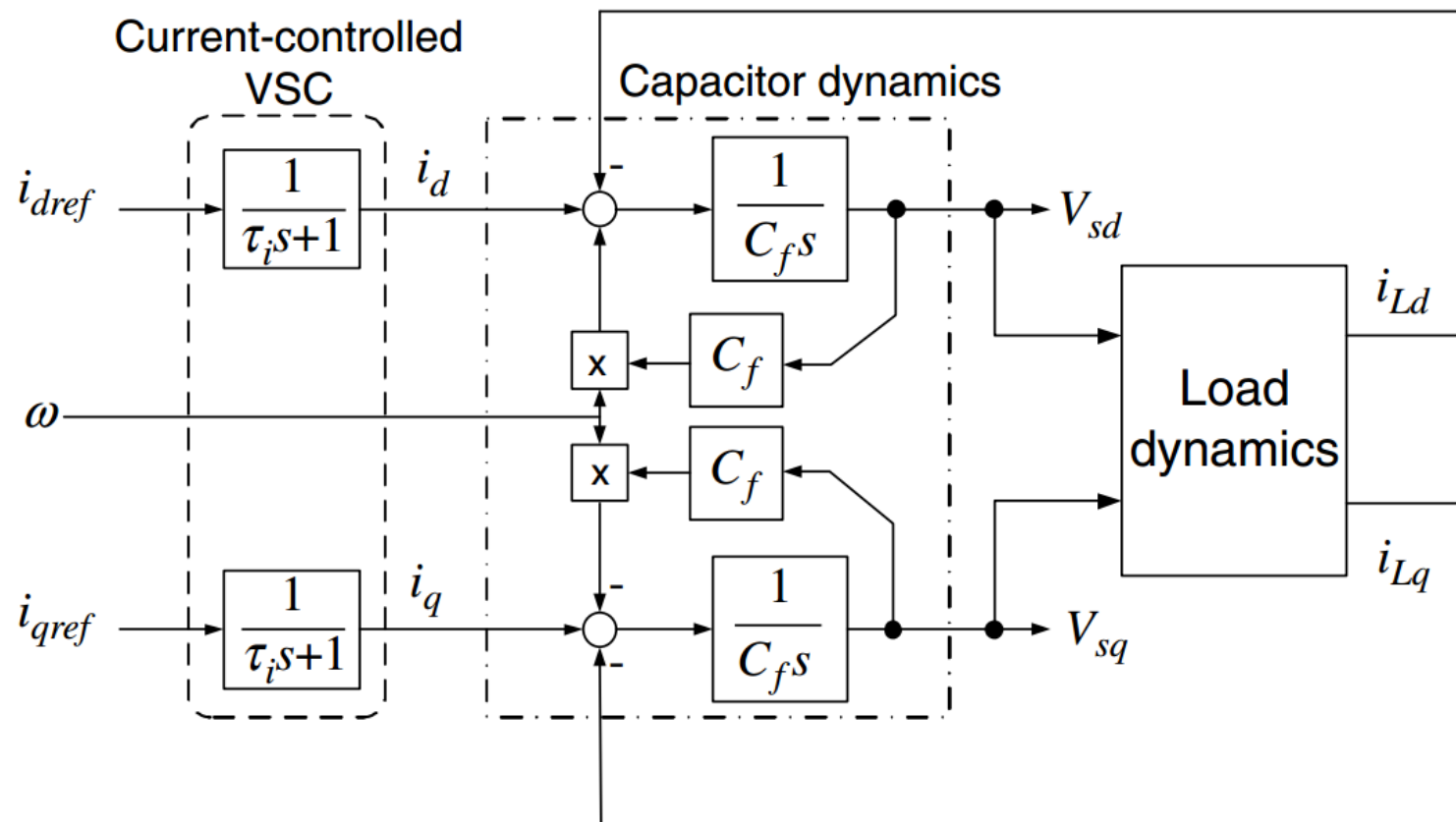
$$f_{dq} = f_d + jf_q = \vec{f}e^{-j\rho}$$

$$\vec{f} = \left(f_d + jf_q \right) e^{j\rho}$$

$$C_f \frac{dv_{sd}}{dt} = i_d - i_{Ld} + \omega C_f v_{sq} \quad C_f \frac{dv_{sq}}{dt} = i_q - i_{Lq} - \omega C_f v_{sd}$$

Grid-Forming Converters

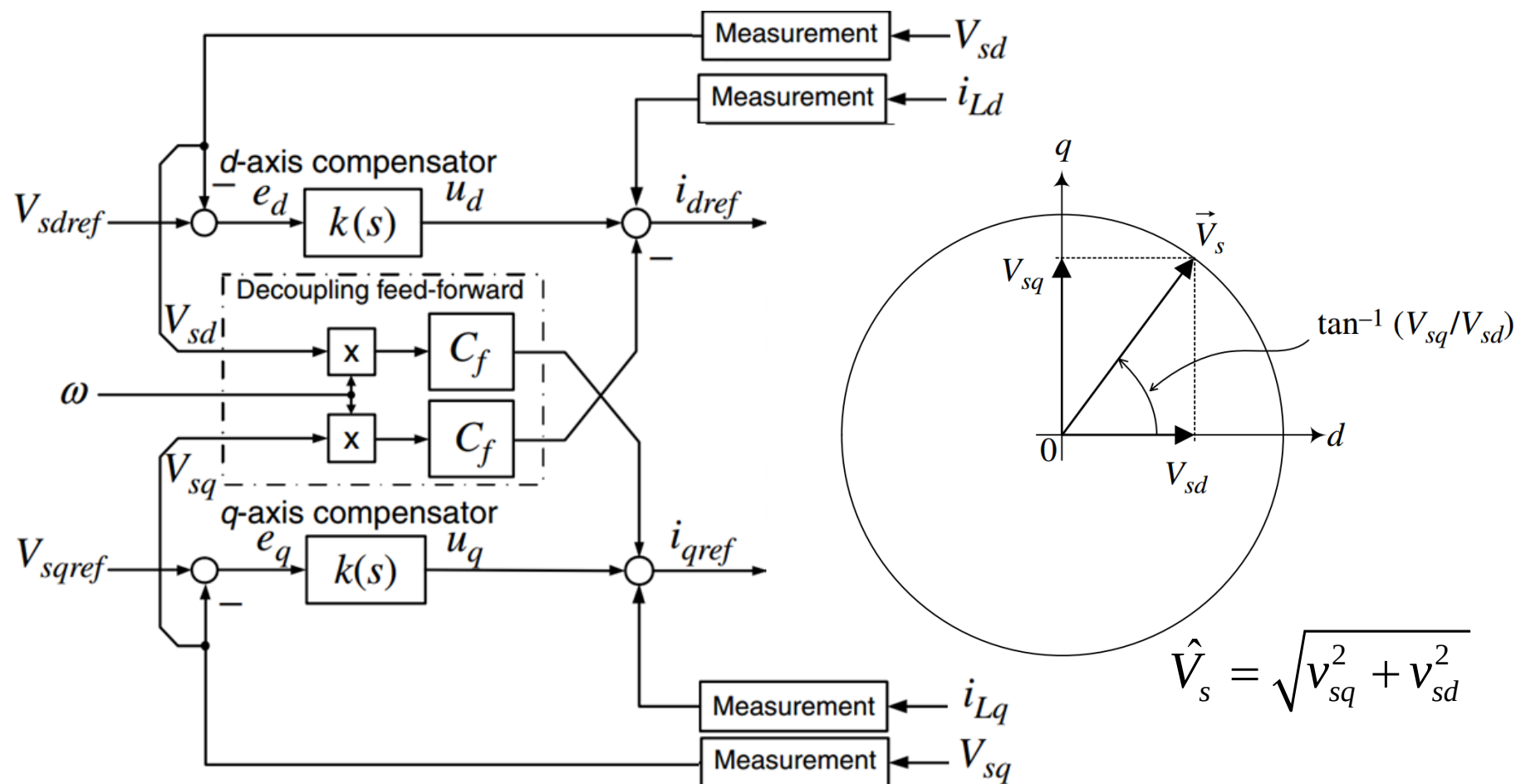
□ Load Voltage Dynamic Model:



$$C_f \frac{dv_{sd}}{dt} = i_d - i_{Ld} + \omega C_f v_{sq} \quad C_f \frac{dv_{sq}}{dt} = i_q - i_{Lq} - \omega C_f v_{sd}$$

Grid-Forming Converters

Load Voltage Control:

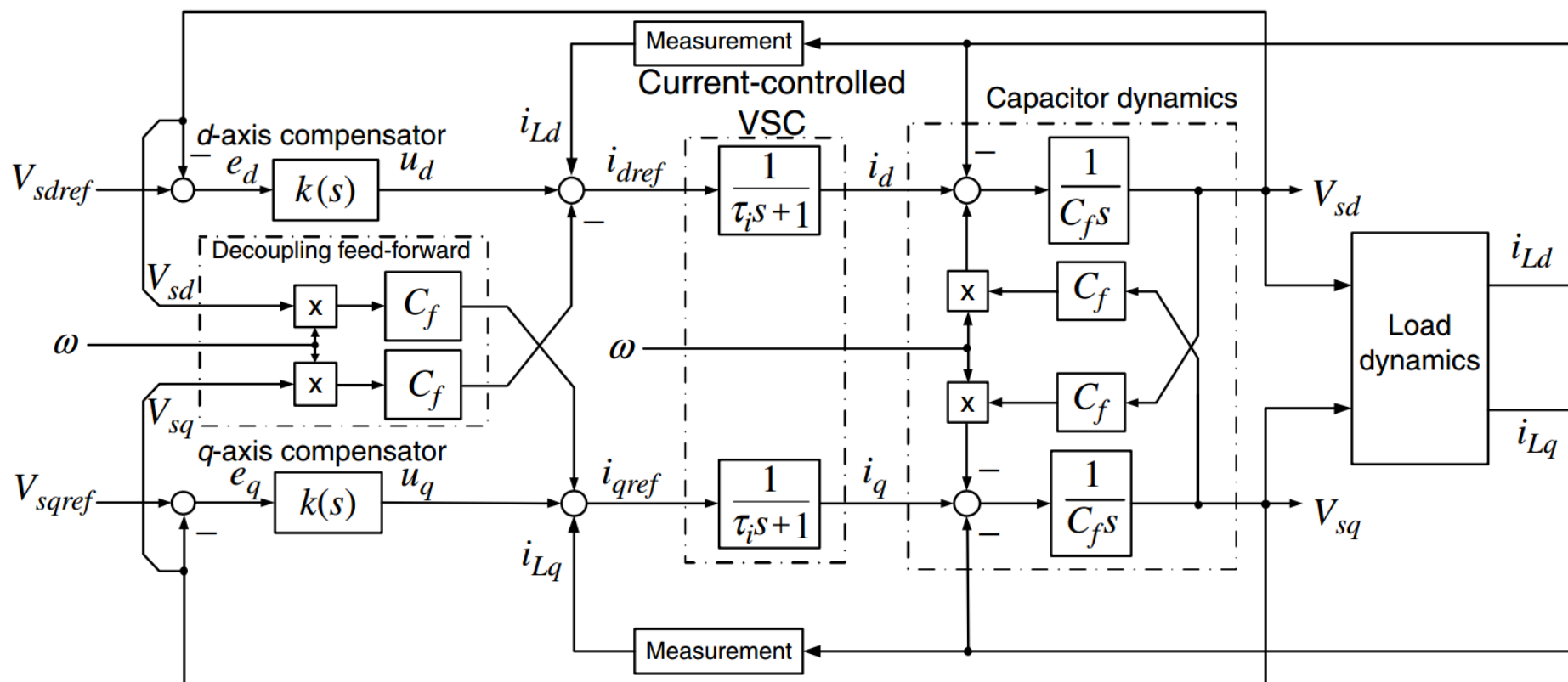


$$i_{dref} = u_d - \omega C_f V_{sq} + i_{Ld}$$

$$i_{qref} = u_q + \omega C_f V_{sd} + i_{Lq}$$

Grid-Forming Converters

Full Control Block Diagram:



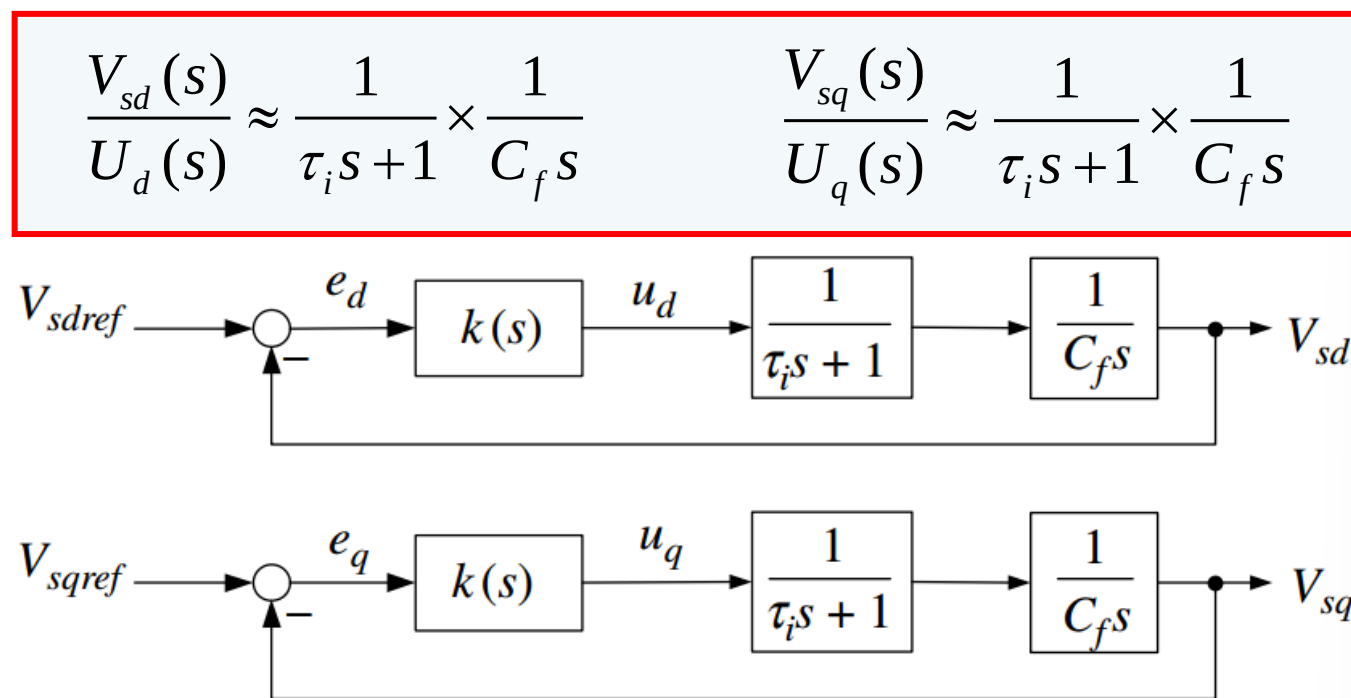
$$\frac{V_{sd}(s)}{U_d(s)} \approx \frac{1}{\tau_i s + 1} \times \frac{1}{C_f s}$$

$$\frac{V_{sq}(s)}{U_q(s)} \approx \frac{1}{\tau_i s + 1} \times \frac{1}{C_f s}$$

Grid-Forming Converters

□ Load Voltage Control:

❖ Compensator Design:



➤ PI Controller:

$$k(s) = k \frac{s+z}{s} \quad \longrightarrow \quad \ell(s) = \left(k \frac{s+z}{s} \right) \left(\frac{1}{1+\tau_i s} \right) \left(\frac{1}{C_f s} \right) = \left(\frac{k}{\tau_i C_f} \right) \left(\frac{s+z}{s+\tau_i^{-1}} \right) \left(\frac{1}{s^2} \right)$$

Grid-Forming Converters

□ Load Voltage Control:

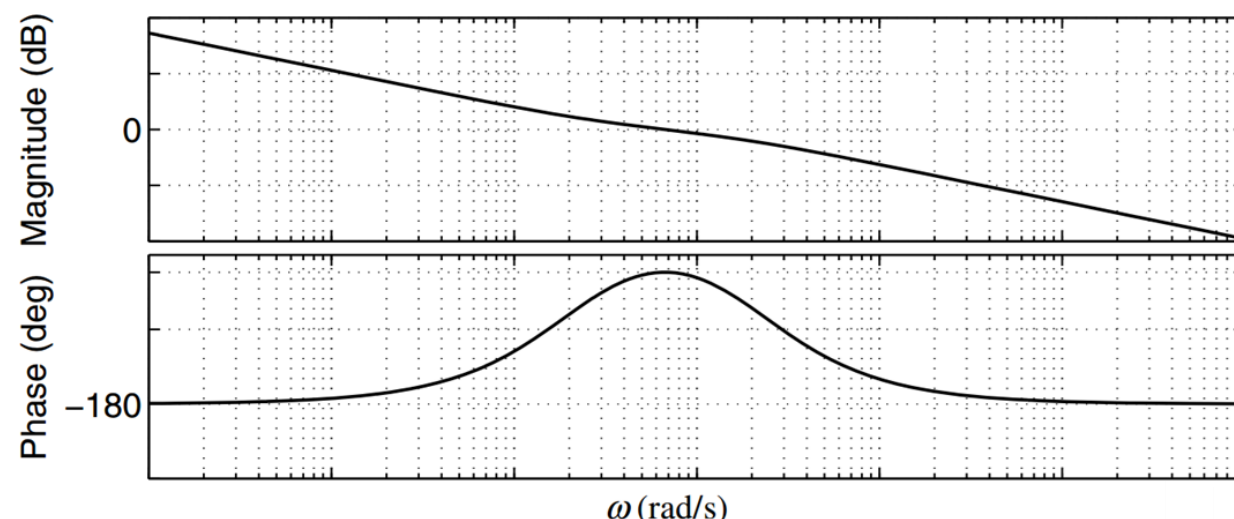
❖ Compensator Design:

Loop Gain:

$$\ell(s) = \left(\frac{k}{\tau_i C_f} \right) \left(\frac{s + z}{s + \tau_i^{-1}} \right) \left(\frac{1}{s^2} \right)$$

➤ PI Controller Parameters:

$$\left\{ \begin{array}{l} \downarrow\downarrow \omega \quad \omega \uparrow\uparrow \\ \omega = \omega_m \end{array} \right. \begin{array}{l} \angle \ell(j\omega) = -180^\circ \\ \angle \ell(j\omega) = -180^\circ + \delta_m \end{array} \xrightarrow{\text{yellow arrow}} \left\{ \begin{array}{l} \omega_m = \sqrt{z/\tau_i} \\ \delta_m = \frac{180^\circ}{\pi} \times \sin^{-1} \left(\frac{1 - \tau_i z}{1 + \tau_i z} \right) \end{array} \right.$$



- If the gain crossover frequency ω_c is chosen as ω_m , then δ_m becomes the phase margin

$$\begin{array}{c} |\ell(j\omega_m)| = 1 \\ \xrightarrow{\text{pink arrow}} k = C_f \omega_m \end{array}$$

Grid-Forming Converters

□ Example: Compensator Design for Voltage Regulation Loop – Example 9.3:

Assume: $\left\{ \begin{array}{lll} R = 2.07 \text{ m}\Omega & L = 100 \text{ }\mu\text{H} & C_f = 2500 \text{ }\mu\text{F} \\ V_{DC} = 1500 \text{ V} & f_{sw} = 1800 \text{ Hz} & \tau_i = 0.5 \text{ ms} \\ V_{sdref} = 400 \text{ V} & V_{sqref} = 0 \text{ V} & \omega_0 = 377 \text{ rad/s} \end{array} \right.$

Desired: Phase margin of 53°

➤ Current Control Loop: $\Rightarrow \left\{ \begin{array}{l} k_p = L/\tau_i = 0.2 \text{ }\Omega \\ k_i = R/\tau_i = 4.14 \text{ }\Omega/\text{s} \end{array} \right. \Rightarrow k_i(s) = 0.2 + \frac{4.14}{s}$

➤ Voltage Control Loop:

$$\delta_m = \frac{180^\circ}{\pi} \times \sin^{-1} \left(\frac{1 - \tau_i z}{1 + \tau_i z} \right) \xrightarrow[\tau_i = 0.5 \text{ ms}]{\delta_m = 53^\circ} z = 224 \text{ rad/s}$$

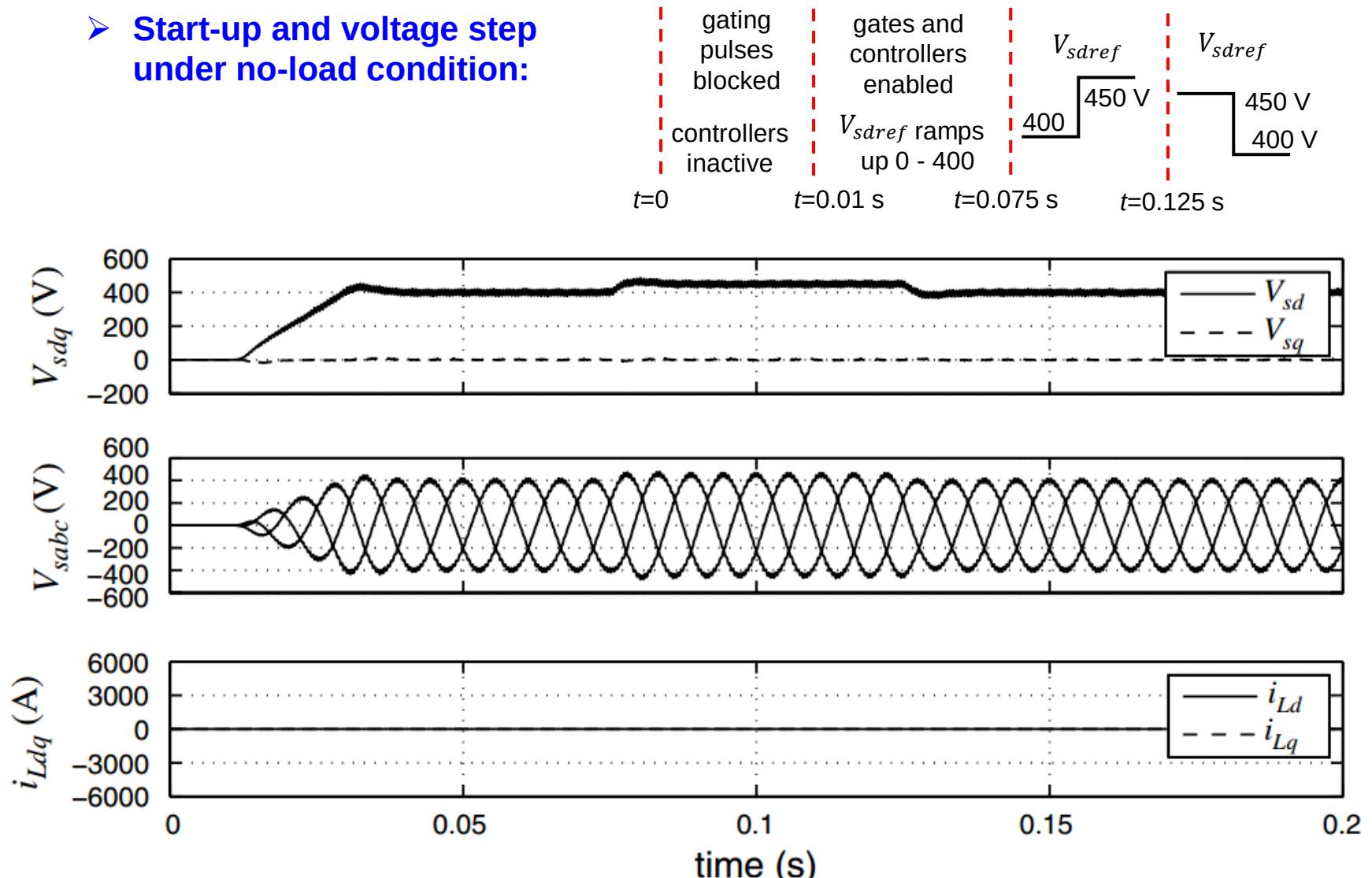
$$\omega_m = \sqrt{z/\tau_i} \xrightarrow[z = 224]{\tau_i = 0.5 \times 10^{-3}} \omega_m = 669 \text{ rad/s}$$

$$k = C_f \omega_m \xrightarrow[\omega_m = 669 \text{ rad/s}]{C_f = 2500 \times 10^{-6}} k = 1.673 \text{ }\Omega^{-1} \Rightarrow k_v(s) = 1.673 \left(\frac{s + 224}{s} \right)$$

Grid-Forming Converters

□ Example: Compensator Design for Voltage Regulation Loop – Example 9.3:

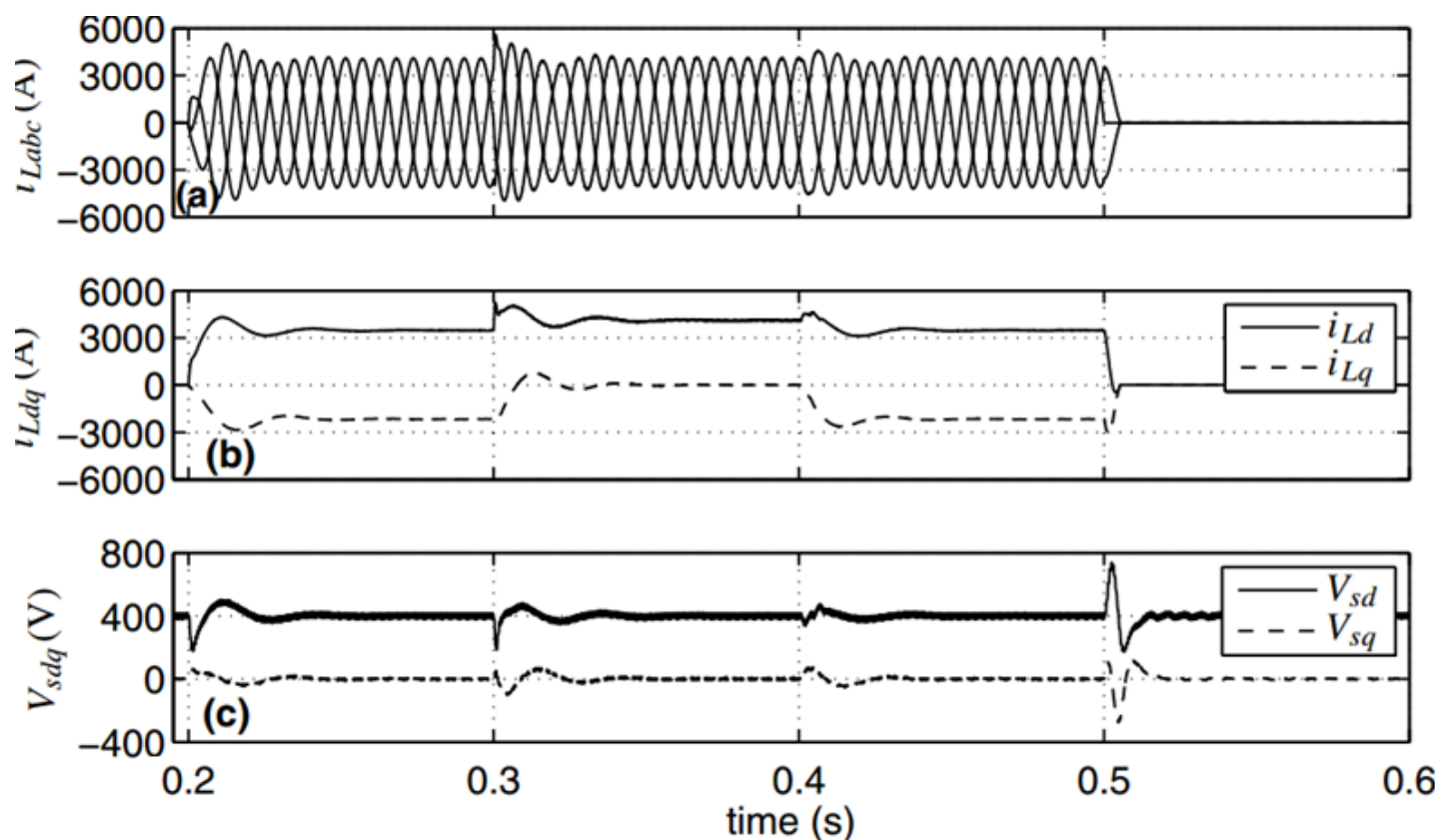
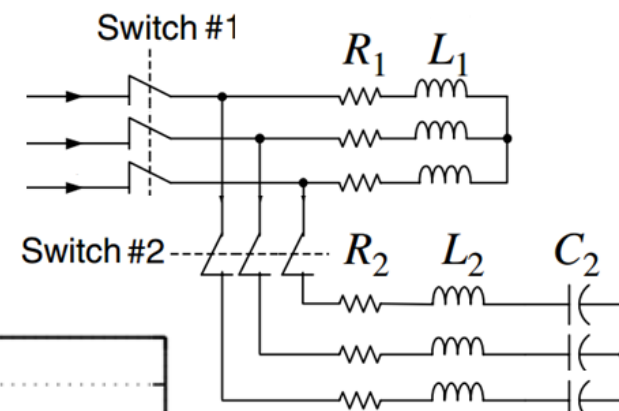
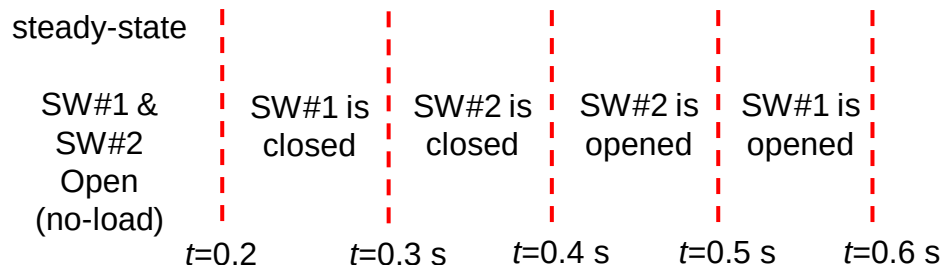
➤ Start-up and voltage step under no-load condition:



Grid-Forming Converters

Example: Compensator Design for Voltage Regulation Loop – Example 9.3:

Load Change:



Load 1:

$$\left\{ \begin{array}{l} R_1 = 83 \text{ m}\Omega \\ L_1 = 137 \text{ }\mu\text{H} \end{array} \right.$$

Load 2:

$$\left\{ \begin{array}{l} R_2 = 50 \text{ m}\Omega \\ L_2 = 68 \text{ }\mu\text{H} \\ C_2 = 13.55 \text{ }\mu\text{F} \end{array} \right.$$

Grid-Forming Converters

Autonomous Operation:

